ADMamba: A Spatio-Temporal Mamba-Inspired Model for Anomaly Detection in Satellite Videos

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***Abstract*—Anomaly detection (AD) has become an increasingly significant research topic for interpreting and understanding remote sensing images. However, most existing studies focus on hyperspectral and high-resolution images, leaving satellite videos relatively unexplored. Compared to these images, AD in satellite videos presents greater challenges due to lower spatial–spectral resolution and the need to process a larger number of frames. In this article, we propose a spatio-temporal Mamba-inspired model (ADMamba) for AD of satellite videos. The proposed ADMamba consists of two main modules: a clustering-based coarse detection module and a Mamba-based fine detection module. First, note that the anomalies are small and distinct from their surroundings but should be detected without any prior knowledge, a clustering-based module is proposed to detect the pseudo-coarse background and anomaly samples. Second, the Mamba-based module is composed of three channels formed from 1-D, 2-D, and 3-D U-Net architectures. This design fully exploits spatial and temporal information, enhancing the feature representation capability of satellite videos. Contrastive losses are adopted to maximize the reconstruction error of pseudo-anomalies while minimizing the reconstruction error of pseudo-backgrounds during the training process. In the testing process, the AD results are obtained by calculating the mean reconstruction error across the three channels. Experiments conducted on three Jilin-1 and two OVS-1 satellite videos demonstrate that the proposed method outperforms the state-of-the-art AD methods. The code for this article is available at https://github.com/hzhdhz/ADMamba**

***Index Terms*—Anomaly detection (AD), decoder, deep learning, encoder, Mamba, satellite videos.**

I. Introduction

VIDEO satellite imagery [1], [2], [3], [4] represents a cutting-edge technology in Earth observation that has rapidly emerged as a prominent area of research due to its capacity to deliver dynamic information. Unlike conventional satellites [5], [6], [7], [8], [9], satellite video systems can capture sequences of images [10], [11], [12], offering detailed insights into moving objects, which is particularly beneficial for applications like vehicle monitoring [13], ship detection [14], and object tracking [15]. Due to its emergence as a new remote sensing technology in recent years, research on satellite videos is still in its early stages. Up to present, the main research directions in satellite videos include super-resolution (SR) [16], [17], [18], [19], [20], target detection [21], [22], [23], [24], object tracking [25], [26], [27], [28], and scene classification [29], [30].

On the other hand, anomaly detection (AD) [31], [32], [33], [34] in remote sensing images is also a significant research focus. The aim of AD is to identify anomalous pixels against the background of land covers [35]. It plays a crucial role in various applications, such as monitoring natural disasters, assessing land use changes, and managing natural resources. In addition, AD can enhance security by recognizing potential threats or illegal activities. Nevertheless, there are several challenges associated with AD, which arise from three main aspects [36], [37], [38]. First, prior information about anomalies and background pixels is not available in advance, which makes both supervised and semi-supervised methods unsuitable for AD. Second, the presence and types of anomalies are unpredictable. The absence of clear patterns or examples makes it challenging to train models effectively. Third, anomalies differ significantly from the background, and their occurrence within the entire scene is relatively low. This results in a severe imbalance between anomalies and the background.

Over the past few decades, significant efforts have been made to address the aforementioned challenges. Existing methods can generally be divided into two categories: traditional methods and deep learning-based methods. One of the widely used traditional methods is the Reed Xiaoli (RX) algorithm [39], which assumes a homogeneous background with a multivariate Gaussian distribution. However, this assumption may not always be sufficient. Subsequently, variants of the RX algorithm (i.e., Global RX (GRX) [40] and local RX (LRX) [41]) are proposed. The GRX is a detector based on the Mahalanobis distance, while LRX is essentially a spatially adaptive version of the GRX statistic that replaces the global background model with a local normal model. A two-step version of the generalized likelihood ratio test (2S-GLRT) method is proposed in [42] to address the multipixel AD problem,

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while fractional Fourier entropy (FrFE) is adopted in [43] to obtain features in an intermediate domain including both spectral and fractional Fourier information. Some approaches also consider AD as a one-class classification issue, which can be addressed using one-class classifiers. For instance, one-class support vector machine (OC-SVM) [44] performs in an unsupervised manner to automatically distinguish the terrain anomalies from normal patterns. In addition, there is a category of advanced traditional approaches based on compressed sensing [45], [46]. These methods assume that background pixels can be expressed through a linear combination of neighboring pixels, while anomalies cannot. The residual from this representation is interpreted as the level of abnormality. For instance, sparse representation (SR) detectors [47], [48], [49] are proposed to represent the pixels being tested by either a manually defined or learned dictionary from the remote sensing data. Although SR-based methods efficiently capture the local structure of the dataset using fewer pixels, they often struggle to adequately represent global structural information. Collaborative representation (CR) detectors [50], [51], [52] enhance the robustness and accuracy of AD results by utilizing the collaborative information from neighboring samples, while low-rank representation (LR) detectors [53], [54], [55], [56] represent the background as a low-rank component and the anomaly as the residual part. Although traditional methods typically have clear mathematical models and physical interpretations, they often require manually defined features or additional assumptions.

In recent years, with the development of artificial intelligence (AI) [57], [58], [59], deep learning-based methods have achieved impressive results in AD for remote sensing images. Among all deep learning methods, convolutional neural networks (CNNs) are the most fundamental. A hybrid CNN and long short-term memory (HCNN-LSTM) [60] is proposed to automatically detect anomalies in farmland using images acquired by the uncrewed aerial vehicles (UAVs). Different CNN architectures, including VGG16, Inception-v3, ResNet-50, and Xception, are utilized for AD of crops in [61], which takes into account the challenges of class imbalance, insufficient variety and quantity of images, and lack of model generalization. An unsupervised learning-based coarse-to-fine multiview network is designed in [62] to perform feature analysis on both global and local attributes within hyperspectral images simultaneously. In addition to CNNs, autoencoders (AEs), U-Nets, and generative adversarial networks (GANs) have also played significant roles in AD. In greater detail, the fundamental idea of AE is to encode the input samples into a lower dimensional space and then decode the original samples from this compressed representation. For instance, an autonomous AD network (Auto-AD) [63] based on AE is proposed to reconstruct the background of hyperspectral images, and the anomalies appear as reconstruction errors. The deep semi-supervised AD (DSAD) [64] is proposed by minimizing the latent distribution entropy for normal data and maximizing it for anomalous data. U-Nets [65] are AEs structured in a "U"-shape, allowing for the transfer of information from intermediate layers of the encoder to the decoder, while GANs [66] can perform discriminative background reconstruction with anomalies being suppressed by learning a new data distribution. In recent years, several self-supervised AD methods have been proposed to learn feature patterns from unlabeled data and identify anomalies based on the deviation of new data from the learned patterns. Among them, blind-spot and blind-block-based methods [67], [68], [69], [70], [71] demonstrate outstanding performance. Transformers [72], [73], which have opened new avenues in various computer vision tasks, also possess considerable potential in AD. The basic idea of using transformers in AD is to leverage their attention mechanisms to capture long-range dependencies in data, allowing for the identification of subtle anomalies across various data modalities. For instance, a convolutional transformer-inspired AE (CTA) is proposed in our previous work [74] to separate the anomalies from the background by tightly mapping the background information in a hypersphere, while a transformer-based spatio-temporal network is proposed in [75] for unsupervised traffic AD of aerial videos. Compared to traditional methods, deep learning-based AD methods can learn hierarchical, abstract, and high-level representations from remote sensing data more effectively. However, both CNNs and transformers have inherent limitations. CNN-based models are constrained by their local receptive fields, which limit their ability to capture long-range dependencies, while transformer-based models excel in global modeling but exhibit quadratic complexity related to the sequence length.

Moreover, a versatile language model backbone named Mamba [76] has been developed since 2023. It builds upon a structured state-space model (SSM) and offers several advantages over transformers.

1) Computational Efficiency: Mamba maintains linear computational complexity, significantly reducing floating-point operations (FLOPs) and memory requirements compared to traditional transformer architectures.

2) Long-Range Dependency Modeling: Mamba excels at capturing long-range dependencies, making it highly effective for tasks that require contextual understanding overextended sequences.

3) Flexible Scanning Mechanisms: Mamba employs bidirectional and zigzag scanning techniques to ensure spatial continuity, enhancing its performance in image processing tasks. Given these advantages, Mamba becomes a state-of-the-art model in deep learning and represents a promising direction for future research. Up to present, Mamba has shown significant advantages in various remote sensing tasks, including image classification [77], [78], [79], [80], semantic segmentation [81], hyperspectral denoising [82], hyperspectral AD [83], [84], change detection [85], [86], SR [87], object tracking [88], and dense prediction [89].

However, to the best of our knowledge, despite the critical importance of AD in satellite videos, up to now, there has been only one conference paper [90] that has explored this topic. Furthermore, although Mamba has shown state-of-the-art performance in multiple domains, its application to AD in satellite video remains unexplored. This represents a significant research gap in remote sensing applications. Notably, AD in satellite videos is more challenging than in hyperspectral or other remote sensing images. This is because satellite videos

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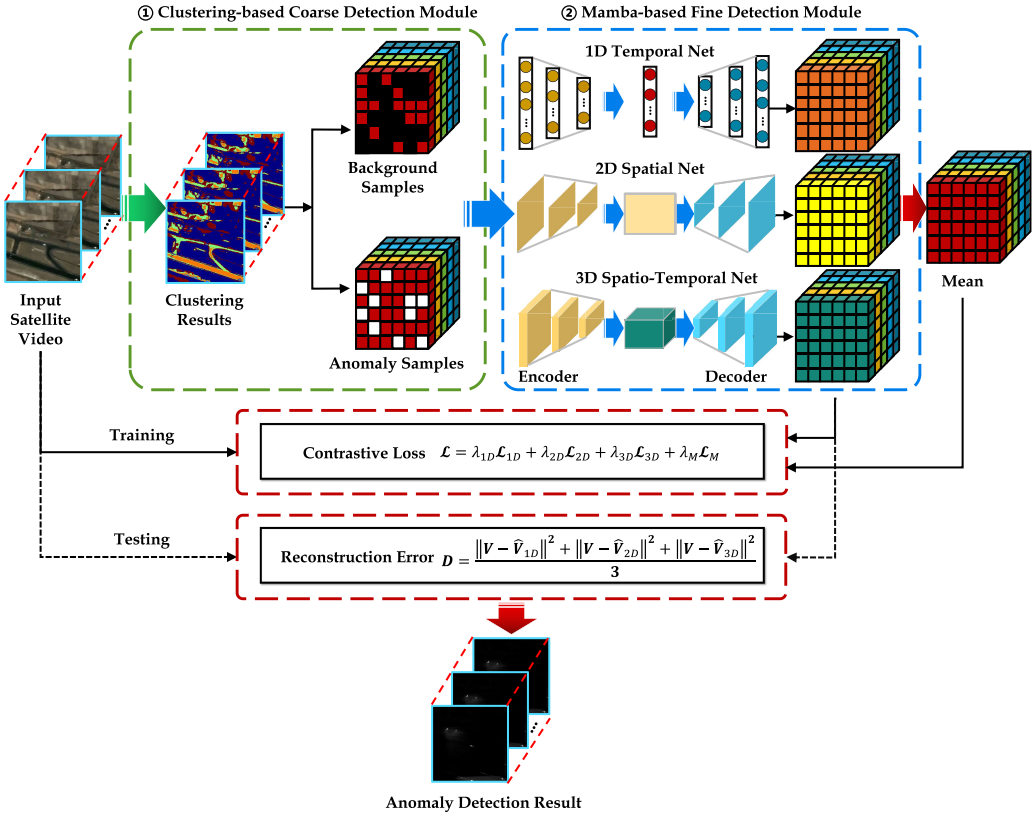


Fig. 1. Overview of the proposed ADMamba.

have lower spatial–spectral resolution and require processing a larger number of frames. In this article, we propose a spatio-temporal **Mamba**-inspired model (ADMamba) for **AD** of satellite videos (see Fig. 1). The ADMamba consists of two main modules: a clustering-based coarse detection module and a Mamba-based fine detection module. The aim of the first module is to detect the pseudo-coarse background and anomaly samples without any prior information regarding anomalies and background pixels, while the latter is composed of three channels with 1-D, 2-D, and 3-D Mamba-based U-Net architectures. The three channels can extract and reconstruct temporal, spatial, and spatio-temporal features by encoder and decoder, respectively, and the final AD results are the mean reconstruction errors of the three channels.

To sum up, the main novelties and contributions of this article are as follows.

1) We are dedicated to AD in satellite videos, which is of crucial importance yet has been severely under-explored so far. Compared to hyperspectral and high-resolution images, satellite videos have lower spatial–spectral resolution but offer more image frames. To address this challenge, an unsupervised AD method is designed to leverage the spatio-temporal information from satellite videos in a coarse-to-fine manner.
2) We propose ADMamba, which, to the best of our knowledge, is the first attempt to develop a Mamba-based deep learning method for AD in satellite videos. Compared to existing convolutional or transformer-based methods, the 1-D, 2-D, and 3-D Mamba-based architectures can more effectively extract and reconstruct the temporal, spatial, and spatio-temporal features of satellite videos.

The rest of this article is organized as follows. In Section II, the proposed ADMamba framework is described in detail. The experimental results and discussions are given in Section III, and Section IV concludes this article.

II. Proposed Method

In this section, we begin with an overview of ADMamba, followed by a detailed description of the architecture of its main modules and the optimization of the model.

A. Overview of the ADMamba

The overview of the proposed ADMamba is depicted in Fig. 1, which inputs the satellite video $\boldsymbol{V} \in \mathbb{R}^{H\times W\times T}$ with T frames $\{\boldsymbol{V}_1, \ldots, \boldsymbol{V}_2, \ldots, \boldsymbol{V}_T\}$ and outputs the AD result $\boldsymbol{D} \in \mathbb{R}^{H\times W\times T}$ with T frames $\{\boldsymbol{D}_1, \ldots, \boldsymbol{D}_2, \ldots, \boldsymbol{D}_T\}$, where H and W are the height and width of the video frame. As shown in Fig. 1, the satellite video $\boldsymbol{V}$ is initially processed through a clustering-based coarse detection module to infer pseudo-background and anomaly samples without any prior information. The motivations of pseudo-label generation are rooted in addressing unavoidable constraints of real-world learning scenarios (i.e., no prelabeled samples are available at the outset for AD in satellite videos) and unique advantages that make the proposed method more practical than alternative solutions for our specific task. Subsequently, we propose a Mamba-based fine detection module that encodes

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$\boldsymbol{V}$ from three perspectives: 1-D temporal, 2-D spatial, and 3-D spatio-temporal. This process generates latent representations in different dimensions, which are then input into three-channel decoders. The decoders aim to reconstruct $\boldsymbol{V}$ using the intermediate latent features from various dimensions. Each channel of this module is a "U"-shape Mamba net. In the training process, contrastive losses are adopted to maximize the reconstruction error of pseudo-anomalies, while the reconstruction error of pseudo-backgrounds is minimized. In the testing process, anomaly detect results are calculated by the mean reconstruction error of the three 1-D, 2-D, and 3-D Mamba-based channels. Crucially, Mamba's implicit recurrence and sequence-modeling capabilities make it particularly suitable for AD in satellite video. This aligns with the fine detection stage, where capturing subtle spatio-temporal patterns is critical for accurate AD.

B. Architecture of Main Modules in the ADMamba

1) Clustering-Based Coarse Detection Module: As shown in Algorithm 1, the coarse detection module takes the satellite video $\boldsymbol{V}$ as input, and outputs the pseudo-background samples $\boldsymbol{S}_{background}$ and anomaly samples $\boldsymbol{S}_{anomaly}$. To this end, $\boldsymbol{V}$ is first stretched into a line $\boldsymbol{V}_{line}$, which can then be used to calculate the Mahalanobis distance $\boldsymbol{M}$

$$\begin{aligned}\boldsymbol{M} &= \boldsymbol{\mathcal{M}}\left(\boldsymbol{V}_{line}, \boldsymbol{V}_{line}\right) \\ &= \sqrt{\left(\boldsymbol{V}_{line}-\boldsymbol{\mu}_V\right)\boldsymbol{\Sigma}^{-1}\left(\boldsymbol{V}_{line}-\boldsymbol{\mu}_V\right)^T}\end{aligned} \tag{1}$$

where $\boldsymbol{\mathcal{M}}(\cdot)$ refers to the Mahalanobis distance function, $\boldsymbol{\mu}_V$ represents the sample mean, $\boldsymbol{\Sigma}$ is the covariance matrix of $\boldsymbol{V}_{line}$. The Mahalanobis distance $\boldsymbol{M}$ quantifies how far $\boldsymbol{V}_{line}$ is from the mean in number of standard deviations.

Afterward, K-means clustering is performed on $\boldsymbol{M}$, and the variance $\boldsymbol{v}_k$ of the kth $(k = 1, 2, \ldots, K)$ cluster is calculated. Taking into account the significant differences between the anomalies and the background, $\boldsymbol{v}_k$ is reordered in descending order to create $\tilde{\boldsymbol{v}}_p, p = 1, 2, \ldots, K$. We then find the samples in $\boldsymbol{V}$ that correspond to $\tilde{\boldsymbol{v}}_p$, represent those samples sequentially as $\tilde{\boldsymbol{V}}_p$, and count the number of samples N_p in $\tilde{\boldsymbol{V}}_p$. Before iteratively obtaining pseudo-anomaly samples and background samples, both $\boldsymbol{S}_{anomaly}$ and $\boldsymbol{S}_{background}$ are initialized as $\emptyset$, and the proportions of anomaly samples $\rho_{anomaly}$ and background samples $\rho_{background}$ in the entire satellite video are set to 0. Subsequently, starting the loop from $p = 1$, the proportion $N_p/(HWT)$ is sequentially added to $\rho_{anomaly}$, $\tilde{\boldsymbol{V}}_p$ is included in $\boldsymbol{S}_{anomaly}$, this continues until $\rho_{anomaly} \geq \rho_1$. Similarly, considering that the variance of the background is generally smaller, the loop begins from $p = K$ and proceeds to 1, sequentially adding the proportion $N_p/(HWT)$ to $\rho_{background}$ and include $\tilde{\boldsymbol{V}}_p$ in $\boldsymbol{S}_{background}$. This continues until $\rho_{background} \geq \rho_2$.

Furthermore, since the number of anomaly samples is generally small, it is necessary to refine $\boldsymbol{S}_{anomaly}$ by removing contiguous samples with an excessive number of anomalies. First, we find the position index $\boldsymbol{I}$ of $\boldsymbol{S}_{\text{anomaly}}$ within $\boldsymbol{V}$. Then, let $\check{\boldsymbol{V}} = \boldsymbol{V} \times 0$, set $\check{\boldsymbol{V}}(\boldsymbol{I}) = 1$, and perform morphological closing on $\check{\boldsymbol{V}}$ to obtain

$$\check{\boldsymbol{V}}_{Close} = \mathcal{I}\left(\check{\boldsymbol{V}}\right) \tag{2}$$

Algorithm 1 Procedure of the Clustering-Based Coarse Detection Module

Input: The satellite video $\boldsymbol{V} \in \mathbb{R}^{H\times W\times T}$
Stretch $\boldsymbol{V}$ into a line $\boldsymbol{V}_{line} \in \mathbb{R}^{(HWT)\times 1}$
Calculate the Mahalanobis distance $\boldsymbol{M}$ by Eq. (1)
Perform K-means clustering on $\boldsymbol{M}$, and calculate the variance $\boldsymbol{v}_k$ of the kth $(k = 1, 2, \ldots, K)$ cluster
Reorder $\boldsymbol{v}_k, k = 1, 2, \ldots, K$ in descending order to create $\tilde{\boldsymbol{v}}_p, p = 1, 2, \ldots, K$
Find the samples $\tilde{\boldsymbol{V}}_p$ in $\boldsymbol{V}$ that correspond to $\tilde{\boldsymbol{v}}_p$
Count the number of samples N_p in $\tilde{\boldsymbol{V}}_p$
Initialize $\boldsymbol{S}_{anomaly} \leftarrow \{\emptyset\}$, $\boldsymbol{S}_{background} \leftarrow \{\emptyset\}$, $\rho_{anomaly} = 0$, and $\rho_{background} = 0$
for $p = 1, 2, \ldots, K$ **do**
  Update $\rho_{anomaly} \leftarrow \rho_{anomaly} + N_p/(HWT)$
  if $\rho_{anomaly} < \rho_1$ **then**
    $\boldsymbol{S}_{anomaly} \leftarrow \{\boldsymbol{S}_{anomaly}, \tilde{\boldsymbol{V}}_p\}$
  else
    Break
  end
end
for $p = K, K-1, \ldots, 1$ **do**
  Update $\rho_{background} \leftarrow \rho_{background} + N_p/(HWT)$
  if $\rho_{background} < \rho_2$ **then**
    $\boldsymbol{S}_{background} \leftarrow \{\boldsymbol{S}_{background}, \tilde{\boldsymbol{V}}_p\}$
  else
    Break
  end
end
Find the position index $\boldsymbol{I}$ of $\boldsymbol{S}_{anomaly}$ in $\boldsymbol{V}$
Let $\check{\boldsymbol{V}} = \boldsymbol{V} \times 0$ and set $\check{\boldsymbol{V}}(\boldsymbol{I}) = 1$, obtain $\check{\boldsymbol{V}}_{Close}$ by Eq. (2)
Identify the connected regions in $\check{\boldsymbol{V}}_{Close}$ and calculate the number of pixels in each connected region
Remove the anomaly samples corresponding to the connected regions in $\check{\boldsymbol{V}}_{Close}$ where the number of pixels is greater than N from $\boldsymbol{S}_{anomaly}$
Output: Pseudo background samples $\boldsymbol{S}_{background}$ and anomaly samples $\boldsymbol{S}_{anomaly}$

where $\mathcal{I}(\cdot)$ is the morphological closing operator, and $\check{\boldsymbol{V}}_{Close}$ is the closing operator result of $\check{\boldsymbol{V}}$. After that, the connected regions within $\check{\boldsymbol{V}}_{Close}$ are identified, and the number of pixels in each of these regions is counted. Finally, when the number of pixels is greater than N, the anomaly samples corresponding to the connected regions in $\check{\boldsymbol{V}}_{Close}$ are removed from $\boldsymbol{S}_{anomaly}$.

2) Mamba-Based Fine Detection Module: After obtaining the pseudo-background samples and anomaly samples, the satellite video $\boldsymbol{V}$ is input into a network consisting of three channels, each comprising an encoder and a decoder structured in a "U"-shape (see Fig. 2). Considering that satellite videos contain rich temporal variation information and spatial texture details, the three channels of this module can, respectively, extract 1-D temporal, 2-D spatial, and 3-D spatio-temporal features, which are then reconstructed by the decoders to form

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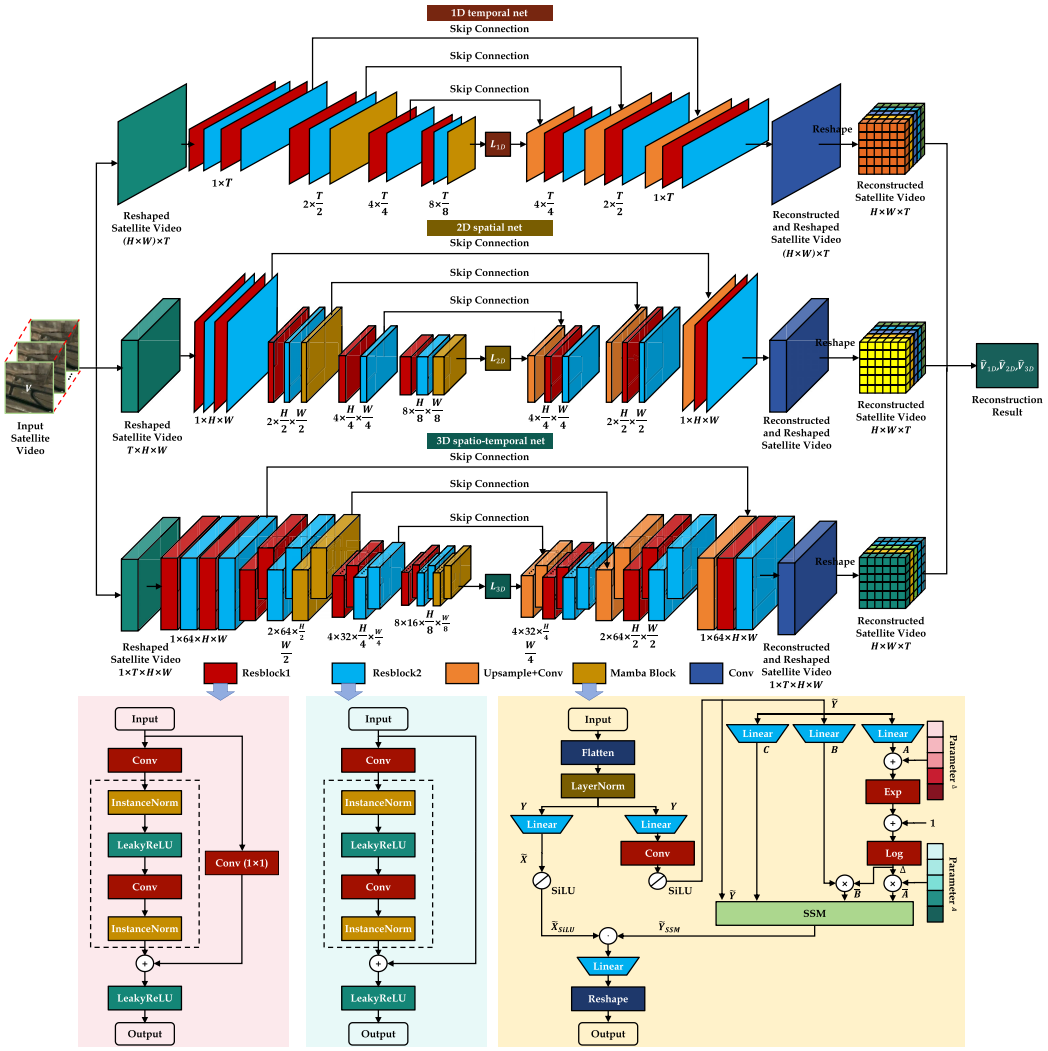


Fig. 2. Architecture of the Mamba-based fine detection module. It is worth noting that although the kernel size of the skip convolution connections in Resblock1 is shown as 1×1, the kernel size is 1 (in the 1-D temporal net) or $1 \times 1 \times 1$ (in the 3-D spatio-temporal net) in the different channels.

$\boldsymbol{V}$. Specifically, the encoders can be expressed as

$$\boldsymbol{L}_{1\text{-}D} = \boldsymbol{\mathcal{F}}_{1\text{-}D\text{-}Encoder}(\boldsymbol{V}, \boldsymbol{\Theta}_{1\text{-}D\text{-}Encoder}) \tag{3}$$

$$\boldsymbol{L}_{2\text{-}D} = \boldsymbol{\mathcal{F}}_{2\text{-}D\text{-}Encoder}(\boldsymbol{V}, \boldsymbol{\Theta}_{2\text{-}D\text{-}Encoder}) \tag{4}$$

$$\boldsymbol{L}_{3\text{-}D} = \boldsymbol{\mathcal{F}}_{3\text{-}D\text{-}Encoder}(\boldsymbol{V}, \boldsymbol{\Theta}_{3\text{-}D\text{-}Encoder}) \tag{5}$$

where $\boldsymbol{\mathcal{F}}_{1\text{-}D\text{-}Encoder}(\cdot)$, $\boldsymbol{\mathcal{F}}_{2\text{-}D\text{-}Encoder}(\cdot)$, and $\boldsymbol{\mathcal{F}}_{3\text{-}D\text{-}Encoder}(\cdot)$ are the encoders in the three channels, $\boldsymbol{\Theta}_{1\text{-}D\text{-}Encoder}$, $\boldsymbol{\Theta}_{2\text{-}D\text{-}Encoder}$, and $\boldsymbol{\Theta}_{3\text{-}D\text{-}Encoder}$ denote the corresponding parameters of various encoders, and $\boldsymbol{L}_{1\text{-}D}$, $\boldsymbol{L}_{2\text{-}D}$, and $\boldsymbol{L}_{3\text{-}D}$ represent the learned latent representation, which can then be fed into the decoders, and the reconstructed videos can be modeled by

$$\hat{\boldsymbol{V}}_{1\text{-}D} = \boldsymbol{\mathcal{F}}_{1\text{-}D\text{-}Decoder}(\boldsymbol{L}_{1\text{-}D}, \boldsymbol{\Theta}_{1\text{-}D\text{-}Decoder}) \tag{6}$$

$$\hat{\boldsymbol{V}}_{2\text{-}D} = \boldsymbol{\mathcal{F}}_{2\text{-}D\text{-}Decoder}(\boldsymbol{L}_{2\text{-}D}, \boldsymbol{\Theta}_{2\text{-}D\text{-}Decoder}) \tag{7}$$

$$\hat{\boldsymbol{V}}_{3\text{-}D} = \boldsymbol{\mathcal{F}}_{3\text{-}D\text{-}Decoder}(\boldsymbol{L}_{3\text{-}D}, \boldsymbol{\Theta}_{3\text{-}D\text{-}Decoder}) \tag{8}$$

where $\boldsymbol{\mathcal{F}}_{1\text{-}D\text{-}Decoder}(\cdot)$, $\boldsymbol{\mathcal{F}}_{2\text{-}D\text{-}Decoder}$, and $\boldsymbol{\mathcal{F}}_{3\text{-}D\text{-}Decoder}$ refer to the decoders in the three channels, $\boldsymbol{\Theta}_{1\text{-}D\text{-}Decoder}$, $\boldsymbol{\Theta}_{2\text{-}D\text{-}Decoder}$, and $\boldsymbol{\Theta}_{3\text{-}D\text{-}Decoder}$ denote the corresponding parameters of various decoders, and $\hat{\boldsymbol{V}}_{1\text{-}D}$, $\hat{\boldsymbol{V}}_{2\text{-}D}$, and $\hat{\boldsymbol{V}}_{3\text{-}D}$ represent the videos reconstructed by different channels.

In greater detail, the architecture of the three channels in ADMamba is illustrated in Fig. 2, from which we can observe that these three channels have similar network structures. Specifically, the architecture of the 1-D temporal net is shown in the first channel of Fig. 2, which takes the reshaped satellite video as input and outputs the reconstructed satellite video. The original frames in $\boldsymbol{V}$ are stretched into vectors, and the vectors corresponding to multiple frames are stacked together to form a matrix of size $(H \times W) \times T$. This matrix is then sequentially input into a 1-D encoder composed of four stages. Stage 1 is a block that consists of two Resblock1 layers and two Resblock2 layers. Both stages 2 and 4 contain a block that includes a Resblock1 layer, a Resblock2 layer, and a Mamba layer, while stage 3 contains a Resblock1 layer and a Resblock2 layer. The number of channels in the four stages are 1, 2, 4, and 8, respectively, while the lengths of the video feature sequences are T, $(T/2)$, $(T/4)$, and $(T/8)$, respectively. Afterward, the output of the encoder (i.e., the latent features learned by the encoder) is fed into a 1-D decoder that also consists of four stages. The first three stages of the decoder have similar structures, which include an upsampling and

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convolution layer, a Resblock1 layer, and a Resblock2 layer. The number of channels in the three stages is 4, 2, and 1, respectively, and the lengths of the video feature sequences are $(T/4)$, $(T/2)$, and T, respectively. Skip connections are used to concatenate the intermediate features from the encoder with the corresponding upsampled features in the decoder. In the fourth stage, a convolution layer is employed to generate the reconstructed and reshaped satellite video. Finally, the output of the fourth stage is reshaped to obtain the final reconstructed satellite video.

The architecture of the 2-D spatial net is illustrated in the second channel of Fig. 2. It reshapes the satellite video $\boldsymbol{V}$ into a format of $T \times H \times W$ and generates the reconstructed satellite video using a 2-D encoder and a 2-D decoder. The 2-D encoder is also composed of four stages, and each stage is similar to the 1-D temporal net. The number of channels of the four stages are 1, 2, 4, and 8, respectively, while feature shapes in each channel vary from $H \times W$, $(H/2) \times (W/2)$, $(H/4) \times (W/4)$ to $(H/8) \times (W/8)$. The 2-D decoder also bears similarities to the 1-D temporal net. The number of channels in the first three stages is 4, 2, and 1, respectively, while the feature shapes are $(H/4) \times (W/4)$, $(H/2) \times (W/2)$, and $H \times W$, respectively. Skip connections serve to preserve high-resolution spatial information from the 2-D encoder and pass it directly to the corresponding 2-D decoder layers. This helps the model recover fine-grained details during the upsampling process. The fourth stage is a convolution layer, which generates a reconstructed and reshaped satellite video with a size of $T \times H \times W$. The final output of the 2-D spatial net is a reconstructed satellite video. In contrast to the 1-D temporal net, the 2-D spatial net focuses more on using 2-D convolutions to extract spatial texture features.

The architecture of the 3-D spatio-temporal net is depicted in the third channel of Fig. 2, which reshapes $\boldsymbol{V}$ into a $1 \times T \times H \times W$ format. The structures of the 3-D encoder and 3-D decoder are analogous to those of the 1-D temporal net and 2-D spatial net. In the 3-D encoder, the number of channels in the four stages is 1, 2, 4, and 8, respectively. The feature depth are 64, 64, 32, and 16, respectively, while the height and width of the features are $H \times W$, $(H/2) \times (W/2)$, $(H/4) \times (W/4)$, and $(H/8) \times (W/8)$, respectively. In the 3D decoder, the number of channels in the former three stages is 4, 2, and 1, respectively. The feature depth recovers from 32 to 64, while the height and width of the features revert from $(H/4) \times (W/4)$ back to $H \times W$. Skip connections are performed to pass high-resolution features from the 3-D encoder to the 3-D decoder, preserving 3-D feature details and improving reconstruction accuracy, especially for fine boundaries and small structures. The fourth stage consists of a 3-D convolution layer that aims to obtain the reconstructed and reshaped satellite video with a size of $1 \times T \times H \times W$. The final output of the 3-D spatio-temporal net is also a reconstructed satellite video. Compared to the 1-D temporal net and the 2-D spatial net, the 3-D spatio-temporal net focuses more on extracting integrated spatio-temporal information.

Moreover, the Mamba block in Fig. 2 takes the learned feature $\boldsymbol{I}_{Mamba}$ as input. The feature $\boldsymbol{I}_{Mamba}$ is then flattened and processed by the layer normalization, which generates $\boldsymbol{Y}$

$$\boldsymbol{Y} = \mathcal{F}_{LN}\left(\mathcal{F}_{Flatten}\left(\boldsymbol{I}_{Mamba}\right)\right) \tag{9}$$

where $\mathcal{F}_{Flatten}(\cdot)$ is the flatten operator and $\mathcal{F}_{LN}(\cdot)$ denotes the layer normalization. Two parallel branches are subsequently adopted to deal with the normalized features. In the first branch, the features are gradually processed by a linear layer, a convolution layer, and a SiLU activation function, whose output is $\tilde{\boldsymbol{Y}}$

$$\tilde{\boldsymbol{Y}} = \mathcal{F}_{SiLU}\left(\mathcal{F}_{Conv}\left(\mathcal{L}_1\left(\boldsymbol{Y}\right)\right)\right) \tag{10}$$

where $\mathcal{L}_1(\cdot)$ and $\mathcal{F}_{Conv}(\cdot)$ are the linear layer and convolution layer, respectively, and $\mathcal{F}_{SiLU}(x) = x * Sigmoid(x)$ is the sigmoid-weighted linear unit activation. Then, $\boldsymbol{A}$, $\boldsymbol{B}$, and $\boldsymbol{C}$ are obtained by performing linear projection on $\tilde{\boldsymbol{Y}}$. Following that, $\boldsymbol{\Delta}$ is generated by applying softplus function to $\boldsymbol{A}$

$$\boldsymbol{\Delta} = \log\left(1 + \exp\left(\boldsymbol{A} + Parameter^{\boldsymbol{\Delta}}\right)\right) \tag{11}$$

where $\boldsymbol{A} = \mathcal{L}_3(\tilde{\boldsymbol{Y}})$, $\mathcal{L}_3(\cdot)$ is the linear layer, $Parameter^{\boldsymbol{\Delta}}$ denotes the learnable parameter. Afterward, $\boldsymbol{\Delta}$ is used to transform the learnable parameter $Parameter^{\boldsymbol{A}}$ and $\boldsymbol{B}$ to $\bar{\boldsymbol{A}}$ and $\bar{\boldsymbol{B}}$, respectively,

$$\bar{\boldsymbol{A}} = \boldsymbol{\Delta} \otimes Parameter^{\boldsymbol{A}} \tag{12}$$

$$\bar{\boldsymbol{B}} = \boldsymbol{\Delta} \otimes \boldsymbol{B} \tag{13}$$

where $\boldsymbol{B} = \mathcal{L}_4(\tilde{\boldsymbol{Y}})$, $\mathcal{L}_4(\cdot)$ is the linear layer, and $\otimes$ refers to matrix multiplication. Next, $\boldsymbol{C}$, $\bar{\boldsymbol{B}}$, and $\bar{\boldsymbol{A}}$ are input into the SSM with $\tilde{\boldsymbol{Y}}$ to obtain the linear modeling result $\tilde{\boldsymbol{Y}}_{SSM}$

$$\tilde{\boldsymbol{Y}}_{SSM} = SSM\left(\boldsymbol{C}, \bar{\boldsymbol{B}}, \bar{\boldsymbol{A}}\right)\left(\tilde{\boldsymbol{Y}}\right) \tag{14}$$

where $\boldsymbol{C} = \mathcal{L}_5(\tilde{\boldsymbol{Y}})$. In the second branch, the feature $\boldsymbol{I}_{Mamba}$ is fed into a linear layer to generate $\tilde{\boldsymbol{X}}$, which is then processed by a SiLU activation function and obtains $\tilde{\boldsymbol{X}}_{SiLU}$

$$\tilde{\boldsymbol{X}}_{SiLU} = \mathcal{F}_{SiLU}\left(\tilde{\boldsymbol{X}}\right) \tag{15}$$

where $\tilde{\boldsymbol{X}} = \mathcal{L}_2(\boldsymbol{Y})$. Finally, $\tilde{\boldsymbol{Y}}_{SSM}$ is multiplied with $\tilde{\boldsymbol{X}}_{SiLU}$, linearly projected, and reshaped to get the final result $\boldsymbol{O}_{Mamba}$

$$\boldsymbol{O}_{Mamba} = \mathcal{F}_{Reshape}\left(\mathcal{L}_6\left(\tilde{\boldsymbol{X}}_{SiLU} \odot \tilde{\boldsymbol{Y}}_{SSM}\right)\right) \tag{16}$$

where $\odot$ denotes dot multiplication, $\mathcal{L}_6(\cdot)$ is the linear layer, and $\mathcal{F}_{Reshape}$ represents the reshape operator.

C. Model Optimization

To optimize the parameters in the Mamba-based fine detection module, the following loss term $\mathcal{L}$ is adopted to constrain the network:

$$\mathcal{L} = \lambda_{1\text{-}D}\mathcal{L}_{1\text{-}D} + \lambda_{2\text{-}D}\mathcal{L}_{2\text{-}D} + \lambda_{3\text{-}D}\mathcal{L}_{3\text{-}D} + \lambda_M\mathcal{L}_M \tag{17}$$

where $\lambda_{1\text{-}D}$, $\lambda_{2\text{-}D}$, $\lambda_{3\text{-}D}$, and λ_M are the weights that balance different losses, $\mathcal{L}_{1\text{-}D}$, $\mathcal{L}_{2\text{-}D}$, $\mathcal{L}_{3\text{-}D}$, and $\mathcal{L}_M$ are the contrastive losses of 1-D temporal, 2-D spatial, 3-D spatio-temporal, and mean reconstruct results, which can be modeled by

$$\begin{aligned}\mathcal{L}_{1\text{-}D} = {} & \mathcal{F}_{CL}\left(\boldsymbol{V}\left(\boldsymbol{P}_{background}\right), \hat{\boldsymbol{V}}_{1\text{-}D}\left(\boldsymbol{P}_{background}\right), 0\right) \\ & + \lambda_{CL\text{-}1\text{-}D}\mathcal{F}_{CL}\left(\boldsymbol{V}\left(\boldsymbol{P}_{anomaly}\right), \hat{\boldsymbol{V}}_{1\text{-}D}\left(\boldsymbol{P}_{anomaly}\right), 1\right)\end{aligned} \tag{18}$$

$$\begin{aligned}\mathcal{L}_{2\text{-}D} = {} & \mathcal{F}_{CL}\left(\boldsymbol{V}\left(\boldsymbol{P}_{background}\right), \hat{\boldsymbol{V}}_{2\text{-}D}\left(\boldsymbol{P}_{background}\right), 0\right) \\ & + \lambda_{CL\text{-}2\text{-}D}\mathcal{F}_{CL}\left(\boldsymbol{V}\left(\boldsymbol{P}_{anomaly}\right), \hat{\boldsymbol{V}}_{2\text{-}D}\left(\boldsymbol{P}_{anomaly}\right), 1\right)\end{aligned} \tag{19}$$

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$$\mathcal{L}_{3\text{-}D} = \mathcal{F}_{CL}\left(\boldsymbol{V}\left(\boldsymbol{P}_{background}\right), \hat{\boldsymbol{V}}_{3\text{-}D}\left(\boldsymbol{P}_{background}\right), 0\right) + \lambda_{CL\text{-}3\text{-}D}\mathcal{F}_{CL}\left(\boldsymbol{V}\left(\boldsymbol{P}_{anomaly}\right), \hat{\boldsymbol{V}}_{3\text{-}D}\left(\boldsymbol{P}_{anomaly}\right), 1\right) \quad (20)$$

$$\mathcal{L}_{M} = \mathcal{F}_{CL}\left(\boldsymbol{V}\left(\boldsymbol{P}_{background}\right), \hat{\boldsymbol{V}}_{M}\left(\boldsymbol{P}_{background}\right), 0\right) + \lambda_{CL\text{-}M}\mathcal{F}_{CL}\left(\boldsymbol{V}\left(\boldsymbol{P}_{anomaly}\right), \hat{\boldsymbol{V}}_{M}\left(\boldsymbol{P}_{anomaly}\right), 1\right) \quad (21)$$

where $\boldsymbol{P}_{background}$ and $\boldsymbol{P}_{anomaly}$ denote the position of $\boldsymbol{S}_{background}$ and $\boldsymbol{S}_{anomaly}$, respectively, $\hat{\boldsymbol{V}}_{M} = ((\hat{\boldsymbol{V}}_{1\text{-}D} + \hat{\boldsymbol{V}}_{2\text{-}D} + \hat{\boldsymbol{V}}_{3\text{-}D})/3)$, $\lambda_{CL\text{-}1\text{-}D}$, $\lambda_{CL\text{-}2\text{-}D}$, $\lambda_{CL\text{-}3\text{-}D}$, and $\lambda_{CL\text{-}M}$ are the weights that balance the loss between background and anomaly samples, $\mathcal{F}_{CL}(\boldsymbol{R}_1, \boldsymbol{R}_2, l)$ is the contrastive loss function, which can be expressed as

$$\mathcal{F}_{CL}\left(\boldsymbol{R}_1, \boldsymbol{R}_2, l\right) = \frac{1-l}{2}\left\|\boldsymbol{R}_1 - \boldsymbol{R}_2\right\|_2^2 + \frac{l}{2}\left(\max\left(0, m - \left\|\boldsymbol{R}_1 - \boldsymbol{R}_2\right\|_2^2\right)\right) \quad (22)$$

where $\boldsymbol{R}_1, \boldsymbol{R}_2$, and l are the arguments of $\mathcal{F}_{CL}$, and $m > 0$ is the margin value. In the function $\mathcal{F}_{CL}$, when the input samples are related to the background samples, setting $l = 0$ minimizes the distance between the satellite video $\boldsymbol{V}$ and the reconstructed videos. In contrast, when the input samples are related to anomaly samples, setting $l = 1$ maximizes the distance between the satellite video $\boldsymbol{V}$ and the reconstructed videos.

By minimizing the loss term $\mathcal{L}$, the Mamba-based fine detection module shown in Fig. 2 becomes proficient at reconstructing background samples, while the reconstruction error for anomaly samples is relatively large. Therefore, during the testing phase, the AD result $\boldsymbol{D}$ can be obtained by calculating the reconstruction error between the original satellite video $\boldsymbol{V}$ and the reconstructed videos $\hat{\boldsymbol{V}}_{1\text{-}D}$, $\hat{\boldsymbol{V}}_{2\text{-}D}$, and $\hat{\boldsymbol{V}}_{3\text{-}D}$

$$\boldsymbol{D} = \frac{\left\|\boldsymbol{V} - \hat{\boldsymbol{V}}_{1\text{-}D}\right\|^2 + \left\|\boldsymbol{V} - \hat{\boldsymbol{V}}_{2\text{-}D}\right\|^2 + \left\|\boldsymbol{V} - \hat{\boldsymbol{V}}_{3\text{-}D}\right\|^2}{3}. \quad (23)$$

III. Experimental Results and Analysis

A. Experimental Datasets

To validate the effectiveness of the proposed ADMamba, experiments are conducted on five satellite videos (see Fig. 3), (i.e., Guadalajara data, Macau data, Bahrain data, Dalian data, and Marseille data). The former three videos are acquired by the Chinese Jilin-1 satellite of Chang Guang Satellite Technology Company Ltd.,[1] while the last two are captured by the Chinese OVS-1 satellite of Zhuhai Orbita Aerospace Science and Technology Company Ltd.[2] All the datasets are cropped into 64 continuous frames with a resolution of 256×256 spatial pixels. Each frame is an RGB image. The details are as follows:

1) *Guadalajara Data:* This data was collected from the Guadalajara International Airport, Mexico. It represents an airport scene that includes runways and airplanes, with five small aircraft considered as anomalies. The Guadalajara data contains a total of 11 652 anomalies, which account for 0.2778% of the entire data, with a total of 4 194 304 pixels.
2) *Macau Data:* This data was collected from the Macau International Airport, China. It is a coastal airport that primarily consists of seawater, a runway, and an airplane. In the experiments, the airplane is considered the anomaly that needs to be detected. This data contains 17 238 anomalies, which account for 0.4110% of the total.
3) *Bahrain Data:* This data was collected from the Bahrain International Airport, Bahrain. The scene includes runways and a plane, with the plane being the one that needs to be detected. The total number of anomalies is 60 186, which accounts for 1.4349% of the entire data.
4) *Dalian Data:* This data was collected from the port of Dalian, China. The scene is a coastal port that includes a dock, navigation channels, berths, breakwaters, harbor facilities, and ships, among which the ship is the anomaly to be detected. The total number of anomalies is 1470, accounting for only 0.0350% of the entire data.
5) *Marseille Data:* This data was collected from the port of Marseille, France. The scene is a port near the sea, mainly including docks, channels, berths, breakwaters, embankments, revetments, storage yards, transportation facilities, harbor facilities, and ships, among which five ships in the water are anomalies to be detected. There are 3364 anomalies in total, which represent only 0.0802% of the whole data.

B. Implementation Details

In the experiments, we compare the ADMamba with various sate-of-the-art methods, including GRX [40], local summation AD based on CR and inverse distance weight (LSAD-CR-IDW) [50], single or multiple local windows and the LR sum-to-one-based AD model (represented as SLW-LRRSTO and MLW-LRRSTO, respectively) [53], 2S-GLRT [42], Auto-AD [63], DSAD [64], CTAnet [74], and blind-spot self-supervised learning network (BS^{3}LNet) [67]. The Auto-AD, DSAD, CTAnet, BS^{3}LNet, and ADMamba are deep learning-based methods, while the others are traditional methods.

All methods are executed on a workstation equipped with dual Intel[3] Xeon3 Gold 5218 CPUs running at 2.30 GHz and an Nvidia GeForce RTX 2080 Ti GPU. The parameters and model structures of the comparison methods are configured to align as closely as possible with the respective References. To promote fairness, all deep learning methods are retrained with the same training samples as those used by ADMamba. The architecture of the ADMamba is shown in Fig. 2, the H, W, and T are set to 256, 256, and 64, respectively. The number of clusters K in the K-means clustering is set to 5. The parameters $\rho_1 = 0.5\%$ and $\rho_2 = 90\%$ are set in Algorithm 1. The weights are set to $\lambda_{CL\text{-}1\text{-}D} = \lambda_{CL\text{-}2\text{-}D} = \lambda_{CL\text{-}3\text{-}D} = \lambda_{CL\text{-}M} = 10$ and $\lambda_{1\text{-}D} = \lambda_{2\text{-}D} = \lambda_{3\text{-}D} = \lambda_M = 1$, respectively. The margin value m in (22) equals 1. Moreover, the Adam solver with $\beta_1 = 0.9$, $\beta_2 = 0.999$ is adopted to train the networks. The learning rate is set to 0.01 and the number of epochs is set to 5000 since

[1] http://charmingglobe.com/
[2] https://www.myorbita.net/index.aspx
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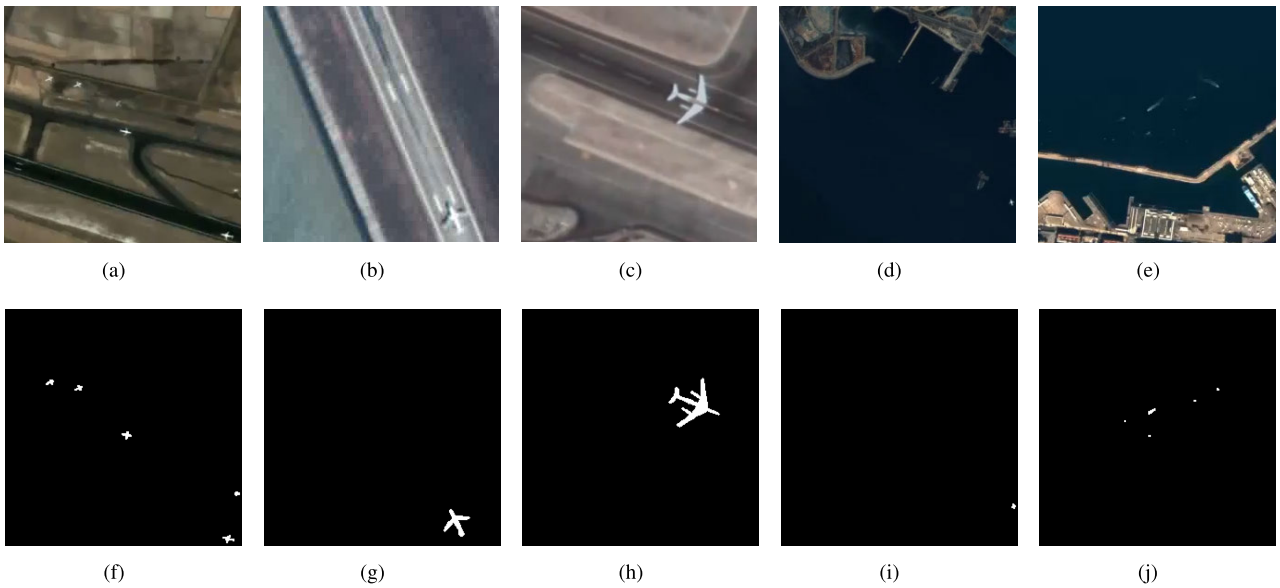


Fig. 3. Illustration of the experimental satellite videos. The first RGB data frame for (a) Guadalajara data, (b) Macau data, (c) Bahrain data, (d) Dalian data, and (e) Marseille data. (f)–(j) Corresponding reference anomaly maps.

further increases in epochs will not yield significantly greater improvements in AD results.

As to the evaluation metrics, the area under the receiver operating characteristic (ROC) curve (AUC) is one of the most widely used index, in which the ROC is a graph illustrating the detection probability P_D against the false alarm probability P_F. In our experiments, multiple different AUC values are used to evaluate the effectiveness of different methods from various perspectives. Specifically, details of the AUC values are as follows.

1) $AUC_{(D,F)}$ is the traditional AUC, which evaluates the effectiveness of a detector.
2) $AUC_{(D,\tau)}$ denotes the target detectability (TD) of a detector.
3) $AUC_{(F,\tau)}$ refers to the AUC along with P_F and τ, which can evaluate the background suppression of a detector.
4) AUC_{TD} is the joint TD of an anomaly detector defined by $AUC_{TD} = AUC_{(D,F)} + AUC_{(D,\tau)} \in [0, 2]$.
5) AUC_{BS} represents the joint BS of an anomaly detector calculated by $AUC_{BS} = AUC_{(D,F)} - AUC_{(F,\tau)} \in [-1, 1]$.
6) AUC_{SNPR} is the signal-to-noise probability ratio (SNPR) given by $AUC_{SNPR} = AUC_{(D,\tau)}/AUC_{(F,\tau)} \in [0, +\infty)$.
7) AUC_{TDBS} is obtained by $AUC_{TDBS} = AUC_{(D,\tau)} - AUC_{(F,\tau)} \in [-1, 1]$.
8) AUC_{ODP} is obtained by $AUC_{ODP} = AUC_{(D,F)} - AUC_{(D,\tau)} \in [-1, 2]$.

C. Visualization of Pseudo-Samples and Learned Features

To gain a more comprehensive understanding of ADMamba, the results of the clustering-based coarse detection module are analyzed, and a detailed examination of the learned features is conducted to shed light on the working mechanism of ADMamba. Without loss of generality, Fig. 4 depicts the results of the clustering-based coarse detection module for the first frame of various datasets. Fig. 5 plots the corresponding error maps for the pseudo-anomaly and pseudo-background samples. It can be seen that the pseudo-anomaly and pseudo-background samples obtained through Algorithm 1 can effectively extract anomaly and background information. Moreover, by comparing the detection results across different datasets, the last two datasets exhibit fewer errors in terms of false positives (i.e., the number of white pixels) than the first three. Specifically, the first three datasets show erroneous pseudo-ADs in the lower middle region, the airplane runway, and the lower right corner, respectively, while the Dalian data has zero false positives and achieves the highest accuracy in detecting pseudo-anomaly samples. To mitigate the risk of biased learning, our approach incorporates specific strategies during the model design, training, and testing phases. The subsequent experimental results demonstrate that these strategies

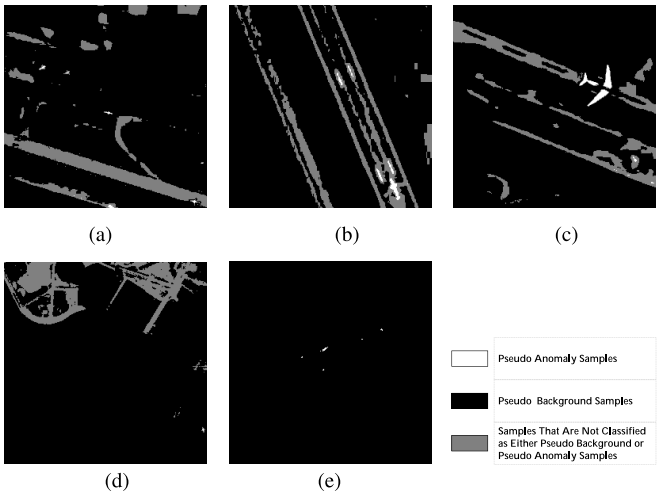


Fig. 4. Results of the clustering-based coarse detection module for the first frame of (a) Guadalajara data, (b) Macau data, (c) Bahrain data, (d) Dalian data, and (e) Marseille data. White, black, and gray represent pseudo-anomaly samples, pseudo-background samples, and samples that are not classified as either pseudo-background or pseudo-anomaly samples, respectively.

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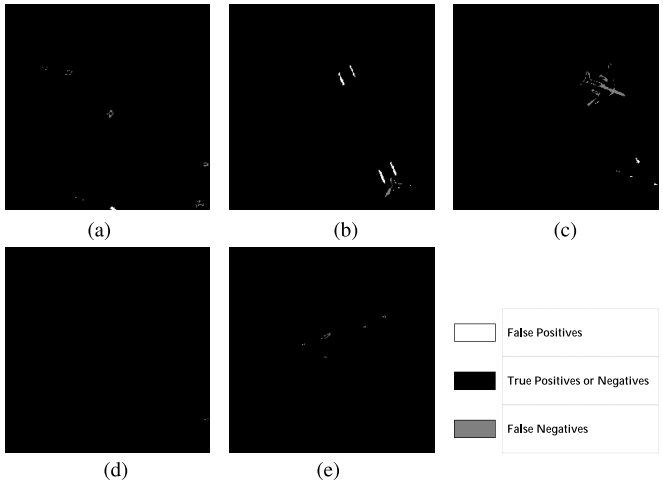


Fig. 5. Error maps of pseudo-samples for the first frame of (a) Guadalajara data, (b) Macau data, (c) Bahrain data, (d) Dalian data, and (e) Marseille data. Ignoring the not classified samples shown in Fig. 4, the samples can be categorized as follows: 1) false positives (white), which are background samples incorrectly detected as pseudo-anomaly samples; 2) true positives or negatives (black), which include both correctly detected background and anomaly samples; and 3) false negatives (gray), which are anomaly samples that are not detected as pseudo-anomaly samples.

reduce the dependence on the accuracy of pseudo-labels and enhance robustness against potential labeling errors throughout the detection process.

To further explore how the Mamba-based fine detection module works, Fig. 6 displays the features extracted by the fourth stage of the 1-D encoder in the 1-D temporal net, the 2-D encoder in the 2-D spatial net, and the 3-D encoder in the 3-D spatio-temporal net for the Guadalajara data. As can be seen, the feature maps demonstrate their ability to capture various structural details, including temporal information, edge and texture information, and spatio-temporal information by utilizing different subnetworks. It is notable that the features learned by the encoders are a distilled, high-level abstract representation of the input satellite video data. They discard redundant details, preserving only the most salient and informative patterns essential for discrimination and decision-making in the downstream AD task. Specifically, Fig. 6(a) shows that most feature representations are broadly consistent, however, one dark-red dashed curve deviates markedly from the others, potentially indicating an abstract representation of anomalous pixels. Fig. 6(b) reveals oblique striations aligned with the runway, this is consistent with Fig. 3(a) and indicates that the 2-D spatial encoder effectively captures edge and texture cues. It preserves scene structure while enhancing directional representations, providing more discriminative features for AD. Fig. 6(c) presents 3-D spatio-temporal features that jointly encode spatial structural properties and temporal dynamics, thereby enhancing the model's understanding of complex scenes and facilitating discrimination between normal background and potential anomalies.

D. AD Accuracy Evaluation and Comparison

The quantitative and qualitative AD results of the competing methods are presented and compared in Tables I and II and

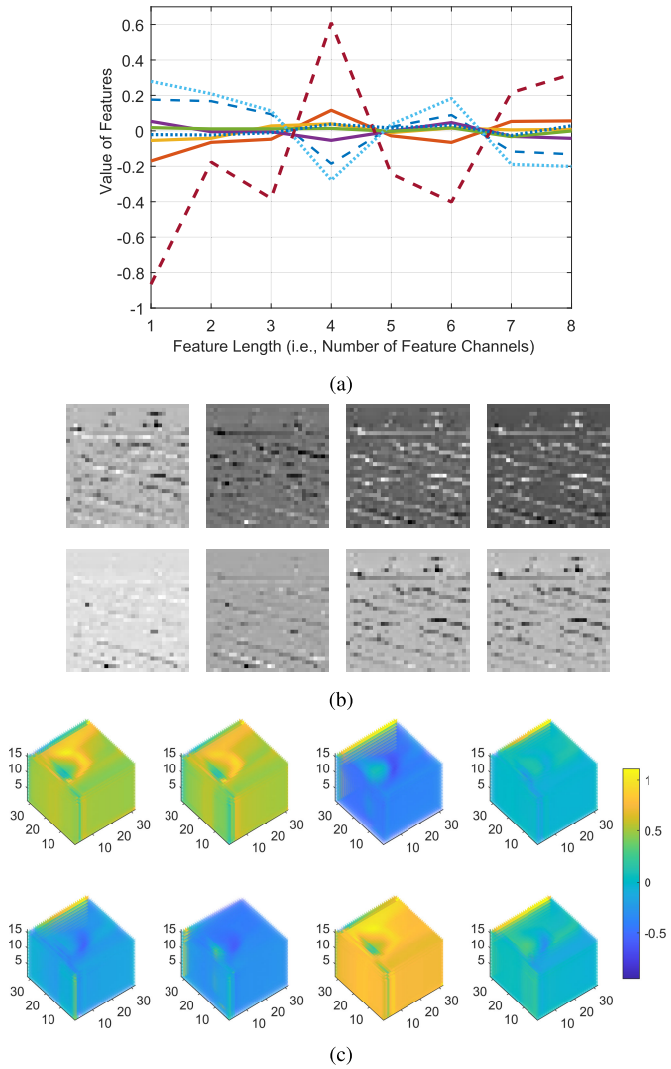


Fig. 6. Diagram displaying the features extracted by the fourth stage of (a) 1-D encoder in the 1-D temporal net, (b) 2-D encoder in the 2-D spatial net, and (c) 3-D encoder in the 3-D spatio-temporal net for the Guadalajara data.

Figs. 7–15. For all the experimental data, various AUC values of the proposed ADMamba are compared against state-of-the-art algorithms in Table I. The best performance is indicated in bold, while the second-best performance is shown in italic underline. Comparison of the $AUC_{(D,F)}$ and its corresponding standard deviations for all frames across different methods is shown in Fig. 7. Without loss of generality, Figs. 8 and 9 show the 3D-ROC curves along with their corresponding 2D-ROC curves for anomaly detectors of various datasets, while Figs. 10–14 display the detection maps for different methods. Moreover, Fig. 15 plots the comparison of total FLOPs and parameters for the proposed ADMamba and deep learning-based methods on the Guadalajara data.

Some observations can be emphasized based on the aforementioned AD results.

1) We found that none of the comparison methods outperformed all others across all datasets and evaluation metrics. This is because the experiments in this article involved multiple datasets and evaluation metrics (i.e., five datasets and eight evaluation metrics), providing a comprehensive and detailed

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TABLE I

AUC OF THE PROPOSED ADMAMBA IS COMPARED AGAINST STATE-OF-THE-ART ALGORITHMS. THE BEST PERFORMANCE IS INDICATED IN BOLD, WHILE THE SECOND-BEST PERFORMANCE IS SHOWN IN ITALIC UNDERLINE

Dataset	Algorithm	$AUC_{(D,F)}\uparrow$	$AUC_{(D,\tau)}\uparrow$	$AUC_{(F,\tau)}\downarrow$	$AUC_{TD}\uparrow$	$AUC_{BS}\uparrow$	$AUC_{SNPR}\uparrow$	$AUC_{TDBS}\uparrow$	$AUC_{ODP}\uparrow$
Guadalajara Data	GRX	0.8404	0.1730	0.0266	1.0134	0.8138	6.5171	*0.1465*	0.9868
	LSAD-CR-IDW	0.5934	**0.4029**	0.2981	0.9963	0.2953	1.3517	0.1048	0.6982
	SLW-LRRSTO	0.6794	0.1356	*0.0019*	0.8149	0.6775	*72.5679*	0.1337	0.8130
	MLW-LRRSTO	*0.8460*	0.2918	0.0581	**1.1378**	0.7878	5.0212	**0.2337**	**1.0797**
	2S-GLRT	0.7022	*0.3071*	0.2238	1.0093	0.4784	1.3724	0.0833	0.7855
	Auto-AD	0.7675	0.0856	0.0028	0.8531	0.7647	31.2551	0.0829	0.8504
	DSAD	0.8406	0.1741	0.0278	*1.0147*	0.8128	6.2684	0.1463	*0.9869*
	CTAnet	0.8437	0.1372	0.0037	0.9809	*0.8400*	36.7857	0.1334	0.9772
	BS^{3}LNet	0.7248	0.0323	0.0033	0.7571	0.7215	9.7973	0.0290	0.7538
	ADMamba	**0.9443**	0.0405	**0.0005**	0.9848	**0.9438**	**93.7887**	0.0400	0.9843
Macau Data	GRX	0.8914	0.4987	0.0575	1.3901	0.8338	8.6688	0.4412	1.3325
	LSAD-CR-IDW	0.9231	**0.7403**	0.3027	**1.6633**	0.6204	2.4460	0.4376	1.3607
	SLW-LRRSTO	0.7415	0.3233	**0.0011**	1.0648	0.7404	**343.8950**	0.3222	1.0637
	MLW-LRRSTO	0.8929	*0.5648*	0.0665	1.4577	0.8264	8.5383	0.4983	1.3912
	2S-GLRT	0.6910	0.5047	0.3874	1.1958	0.3036	1.3028	0.1173	0.8083
	Auto-AD	*0.9685*	0.3280	*0.0031*	1.2965	**0.9654**	*111.6050*	0.3249	1.2934
	DSAD	0.8928	0.5000	0.0570	1.3929	0.8359	8.7787	0.4430	1.3359
	CTAnet	0.9433	0.5247	0.0047	1.4680	0.9386	111.4194	**0.5200**	*1.4633*
	BS^{3}LNet	0.6931	0.1777	0.0518	0.8709	0.6413	4.4387	0.1259	0.8190
	ADMamba	**0.9720**	0.5433	0.0251	*1.5153*	*0.9469*	21.7014	*0.5182*	**1.4902**
Bahrain Data	GRX	0.8057	0.4507	0.1010	1.2564	0.7047	4.4630	0.3497	1.1554
	LSAD-CR-IDW	*0.8846*	**0.7401**	0.3763	**1.6247**	0.5083	1.9670	0.3638	*1.2484*
	SLW-LRRSTO	0.8157	*0.5612*	0.1894	1.3769	0.6264	2.9637	0.3718	1.1875
	MLW-LRRSTO	0.8218	0.4589	0.0496	1.2807	0.7722	9.2935	*0.4093*	1.2311
	2S-GLRT	0.7865	0.5054	0.3370	1.2919	0.4495	1.5007	0.1684	0.9549
	Auto-AD	0.8466	0.0811	**0.0054**	0.9277	*0.8412*	15.9343	0.0756	0.9223
	DSAD	0.8004	0.4480	0.1034	1.2484	0.6969	4.3327	0.3446	1.1450
	CTAnet	0.8423	0.3909	0.0235	1.2331	0.8187	*16.5995*	0.3673	1.2096
	BS^{3}LNet	0.7582	0.0626	*0.0103*	0.8208	0.7478	8.5241	0.0523	0.8105
	ADMamba	**0.9935**	0.5528	0.0148	*1.5463*	**0.9787**	**37.7531**	**0.5381**	**1.5315**
Dalian Data	GRX	0.8813	0.2409	0.0101	1.1223	0.8712	23.7638	0.2308	1.1121
	LSAD-CR-IDW	0.9452	*0.4373*	0.0706	*1.3825*	0.8745	6.1930	*0.3667*	*1.3118*
	SLW-LRRSTO	0.8715	0.3242	0.0155	1.1956	0.8559	20.8636	0.3086	1.1801
	MLW-LRRSTO	0.8897	0.3743	0.0345	1.2640	0.8552	10.8441	0.3398	1.2295
	2S-GLRT	0.9759	**0.8159**	0.0282	**1.7918**	0.9478	28.8551	**0.7877**	**1.7637**
	Auto-AD	0.8981	0.0991	0.0023	0.9972	0.8958	42.7279	0.0968	0.9949
	DSAD	0.6335	0.2109	0.0268	0.8444	0.6067	7.8597	0.1841	0.8176
	CTAnet	0.8743	0.2164	0.0035	1.0907	0.8709	62.4827	0.2129	1.0873
	BS^{3}LNet	**0.9852**	0.2245	**0.0001**	1.2097	**0.9851**	**2743.1998**	0.2244	1.2096
	ADMamba	*0.9847*	0.2473	*0.0008*	1.2320	*0.9839*	*343.5452*	0.2465	1.2312
Marseille Data	GRX	0.5933	0.0462	0.0482	0.6395	0.5451	0.9591	-0.0020	0.5913
	LSAD-CR-IDW	0.8315	0.2682	0.1245	1.0998	0.7071	2.1587	0.1438	0.9753
	SLW-LRRSTO	0.4825	0.0007	0.0137	0.4833	0.4689	0.0523	-0.0130	0.4696
	MLW-LRRSTO	0.6957	0.0902	0.0604	0.7859	0.6353	1.4954	0.0298	0.7255
	2S-GLRT	0.8947	0.2402	0.0802	1.1349	0.8144	3.0089	0.1600	1.0547
	Auto-AD	*0.8980*	0.1227	0.0223	1.0208	*0.8757*	5.5099	0.1005	0.9985
	DSAD	0.8089	0.1077	0.0466	0.9166	0.7623	2.3035	0.0611	0.8700
	CTAnet	0.8955	*0.3090*	0.1039	*1.2045*	0.7916	2.9732	*0.2050*	*1.1005*
	BS^{3}LNet	0.8756	0.0840	*0.0041*	0.9597	0.8715	*23.1581*	0.0800	0.9556
	ADMamba	**0.9782**	**0.3920**	**0.0003**	**1.3702**	**0.9779**	**1592.9631**	**0.3917**	**1.3700**

assessment of the different methods. Given the large number of datasets and evaluation metrics, it is difficult for any single method to achieve the best performance in every aspect. This further underlines the fact that research on AD in satellite videos remains in its early stages, thus justifying the need for more extensive attention and in-depth investigations in the future.

2) GRX performs well on most datasets. For instance, on the four datasets other than the Marseille data, at least two methods perform worse than GRX across most evaluation metrics.

3) Except for the Guadalajara data, LSAD-CR-IDW outperforms methods like GRX, SLW-LRRSTO, and MLW-LRRSTO in detection performance across most datasets. For instance, on the Bahrain data, LSAD-CR-IDW achieves the best or second-best results on four metrics.

4) In most cases, the MLW-LRRSTO outperforms the SLW-LRRSTO. For example, on the Macau data, the $AUC_{(D,F)}$, $AUC_{(D,\tau)}$, AUC_{TD}, AUC_{BS}, AUC_{TDBS}, and AUC_{ODP} for MLW-LRRSTO are higher than those for SLW-LRRSTO by 0.1514, 0.2415, 0.3929, 0.0860, 0.1761, and 0.3275, respectively. However, in most cases, MLW-LRRSTO has weaker

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TABLE II

COMPUTATION TIME (FOR DEEP LEARNING-BASED METHODS, COMPUTATION TIME IS TRAINING/TESTING TIME; THE UNIT OF TIME IS SECONDS) OF DIFFERENT METHODS

Algorithm	Guadalajara Data	Macau Data	Bahrain Data	Dalian Data	Marseille Data
GRX	2.5965	2.1369	2.1498	2.1384	2.1355
LSAD-CR-IDW	674.3455	671.5736	673.0690	676.2630	674.4292
SLW-LRRSTO	13,714.5839	13,434.3015	13,317.7938	13,381.8633	13,332.9133
MLW-LRRSTO	22,701.5042	22,742.9861	22,591.5198	21,175.2356	21,245.3205
2S-GLRT	128,355.7908	120,799.9154	117,609.1557	118,785.9817	117,200.6059
Auto-AD	11.4644/0.0558	11.9039/0.0582	11.7302/0.0578	11.6113/0.0553	11.4506/0.0561
DSAD	4,529.9951/194.6850	4,596.1711/196.1575	4,585.8418/194.1571	4,480.8458/188.7355	4,350.5417/184.5412
CTAnet	226,801.2635/457.2541	224,181.5474/453.5549	224,125.5878/452.2578	225,123.2547/454.8745	227,424.5931/458.4784
BS^{3}LNet	33,528.3732/0.4218	33,287.2514/0.4123	32,336.2231/0.4054	34,121.1311/0.4663	34,019.8751/0.4601
ADMamba	61,975.5412/2.5741	61,260.8745/2.4187	60,475.2501/2.3402	62,975.8474/2.5412	62,460.4741/2.4502

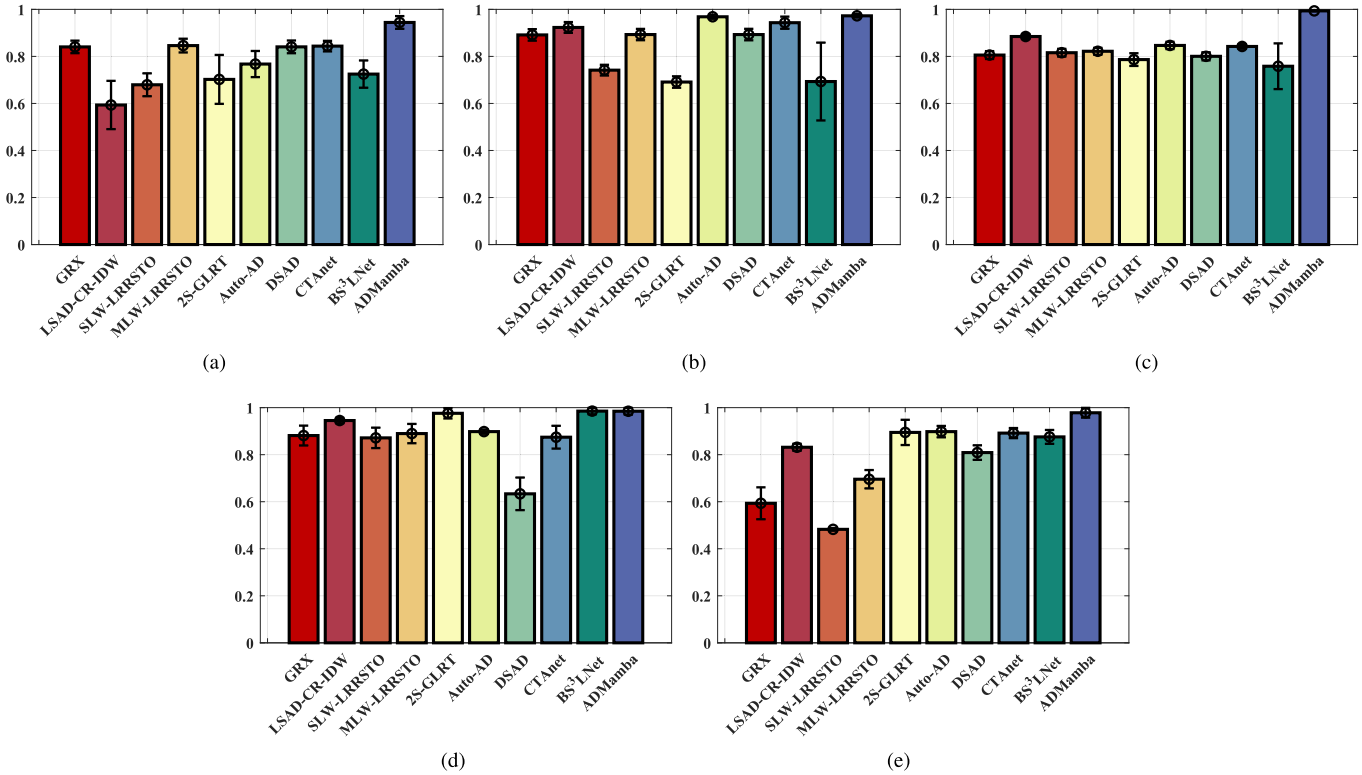


Fig. 7. Comparison of the $AUC_{(D,F)}$ and its corresponding standard deviations across different methods. (a) Guadalajara data. (b) Macau data. (c) Bahrain data. (d) Dalian data. (e) Marseille data.

background suppression and lower detection results in terms of SNPR compared to SLW-LRRSTO. For instance, on the Macau data, SLW-LRRSTO outperforms MLW-LRRSTO in $AUC_{(F,\tau)}$ and AUC_{SNPR} by 0.0654 and 335.3567, respectively. Similar trends are observed across other datasets.

5) When the proportion of anomalies is extremely low, the detection performance of the 2S-GLRT method becomes more prominent. For instance, in the Dalian and Marseille datasets, where the anomaly proportions are only 0.0350% and 0.0802%, respectively, the 2S-GLRT outperforms GRX, LSAD-CR-IDW, SLW-LRRSTO, MLW-LRRSTO, and DSAD across most metrics. Specifically, in the Dalian data, the $AUC_{(D,F)}$ of 2S-GLRT is improved by 0.0946, 0.0307, 0.1044, 0.0862, and 0.3424 compared to GRX, LSAD-CR-IDW, SLW-LRRSTO, MLW-LRRSTO, and DSAD, respectively.

6) Deep learning-based methods do not always outperform traditional methods across all datasets and evaluation metrics. For example, in the Macau data, the $AUC_{(D,F)}$ of the DSAD is 0.0001 lower than that of the MLW-LRRSTO. In the Bahrain data, the $AUC_{(D,\tau)}$ of the Auto-AD is 0.3778 lower than that of MLW-LRRSTO. However, the $AUC_{(F,\tau)}$ of the Auto-AD is improved by 0.0442 compared to MLW-LRRSTO. There are two main reasons why deep learning does not always work better than traditional methods. First, traditional methods, having been refined over many years, are specifically designed for AD. Second, deep learning encounters significant challenges in unsupervised scenarios due to the lack of labeled data for learning.

7) In most datasets, the Auto-AD outperforms the DSAD in terms of both $AUC_{(D,F)}$ and $AUC_{(F,\tau)}$ metrics. However, for

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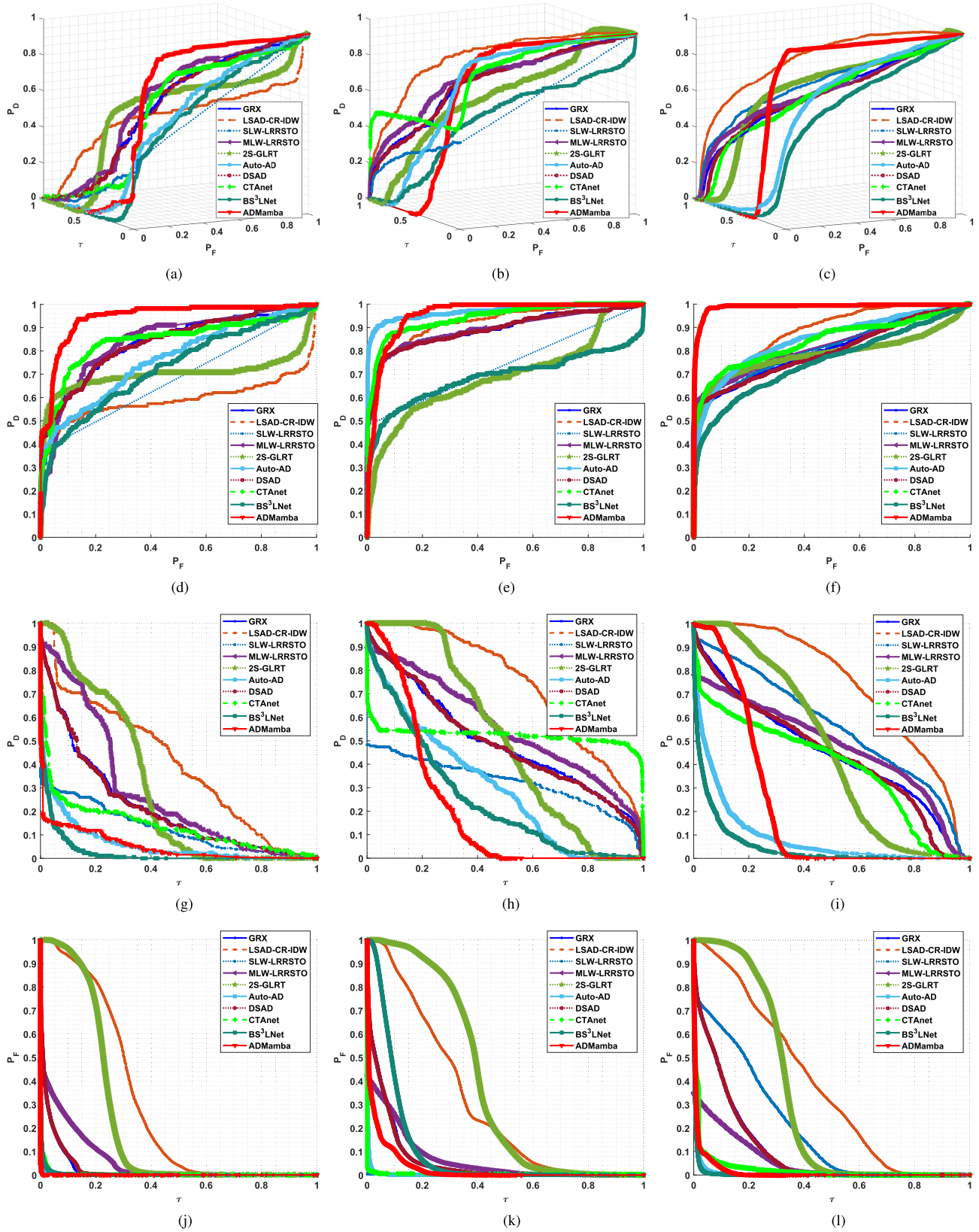


Fig. 8. 3D-ROC curves along with their corresponding 2D-ROC curves for anomaly detectors of the Guadalajara data (see the first column), Macau data (see the second column), and Bahrain data (see the third column). (a)–(c) 3D-ROC curves, (d)–(f) 2D-ROC curves of (P_D, P_F), (g)–(i) 2D-ROC curves of (P_D, τ), and (j)–(l) 2D-ROC curves of (P_F, τ). P_D, P_F, and τ are the detection probability, the false alarm probability, and the threshold, respectively.

other indicators, the experimental results are generally lower than those of DSAD. For instance, on the Bahrain dataset, Auto-AD improves $AUC_{(D,F)}$ and $AUC_{(F,\tau)}$ by 0.0462 and 0.0980, respectively, compared to DSAD. On the other hand,

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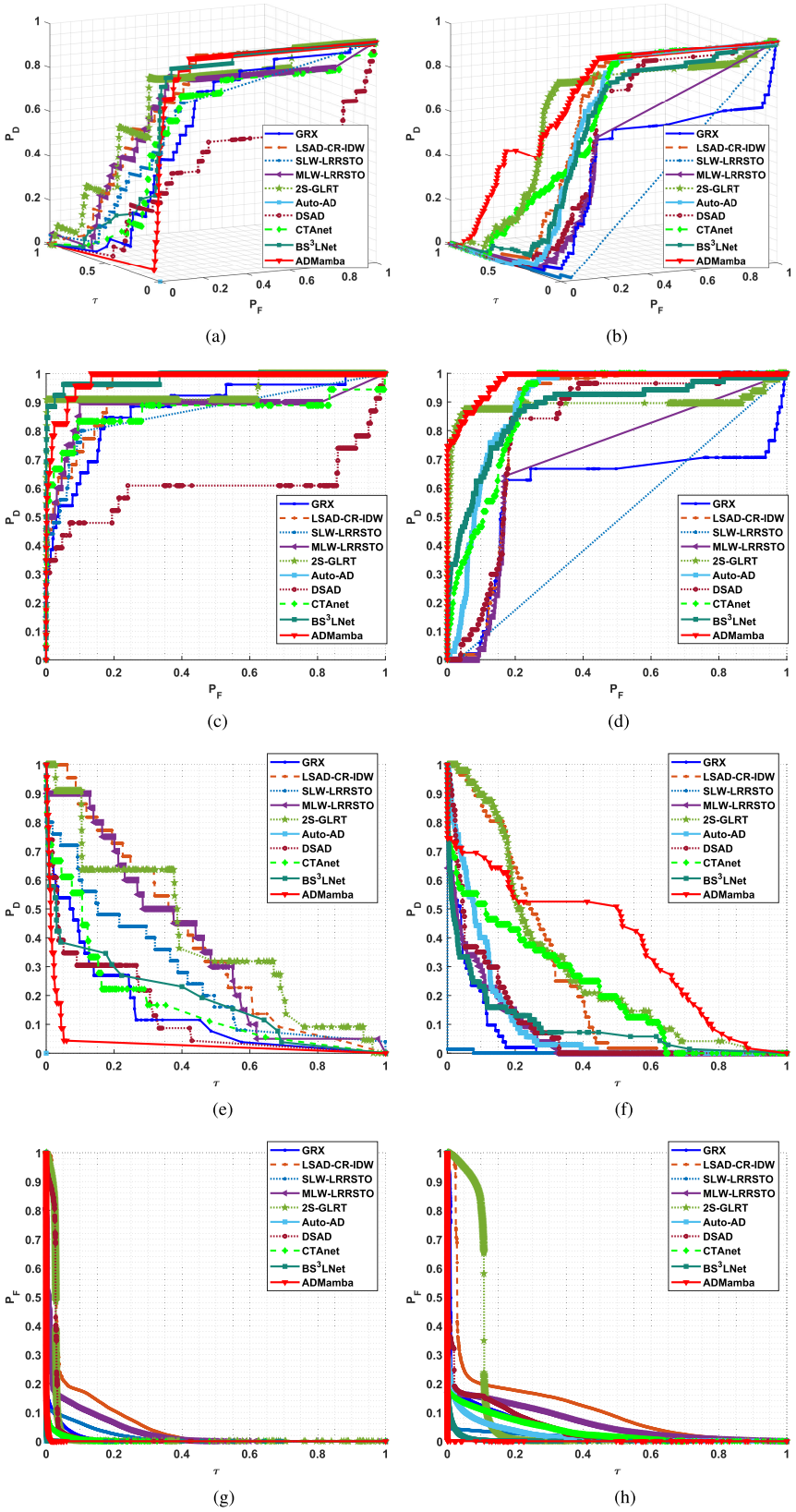


Fig. 9. 3D-ROC curves along with their corresponding 2D-ROC curves for anomaly detectors of both the Dalian data (see the first column) and the Marseille data (see the second column). (a) and (b) 3D-ROC curves, (c) and (d) 2D-ROC curves of (P_D, P_F), (e) and (f) 2D-ROC curves of (P_D, τ), and (g) and (h) 2D-ROC curves of (P_F, τ). P_D, P_F, and τ are the detection probability, the false alarm probability, and the threshold, respectively.

Auto-AD shows a decrease of 0.3669 in $AUC_{(D,\tau)}$ and 0.3207 in AUC_{TD} compared to DSAD.

8) CTAnet shows good performance and is relatively stable across all datasets. For example, CTAnet sometimes

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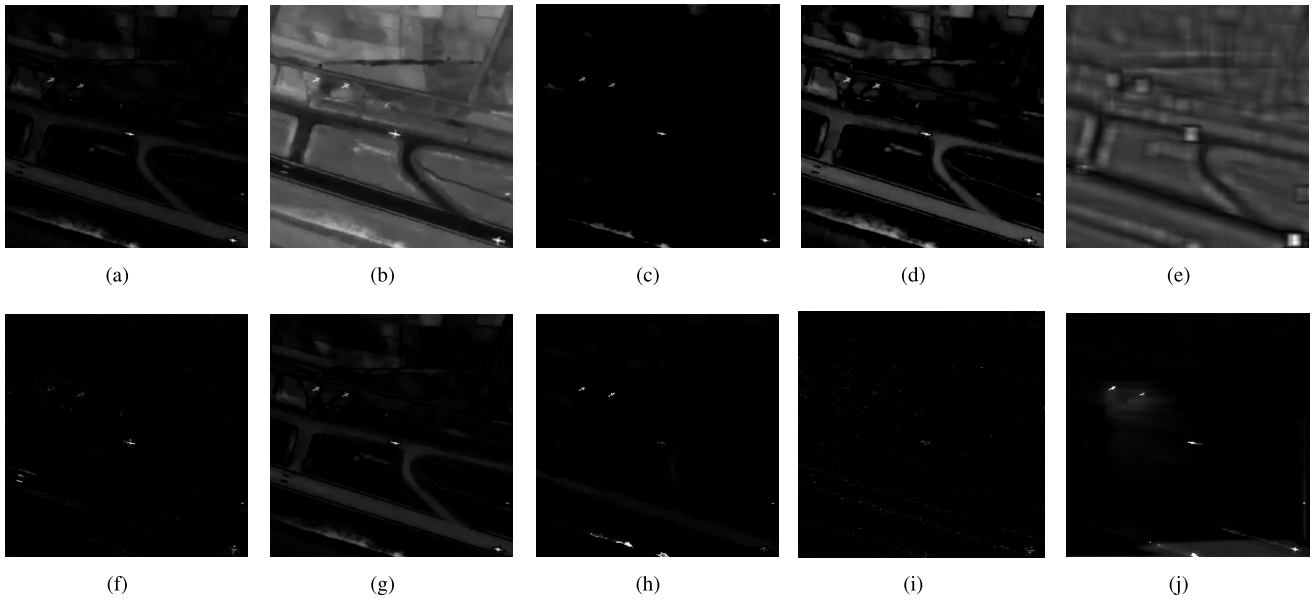


Fig. 10. Detection maps of (a) GRX, (b) LSAD-CR-IDW, (c) SLW-LRRSTO, (d) MLW-LRRSTO, (e) 2S-GLRT, (f) Auto-AD, (g) DSAD, (h) CTAnet, (i) BS^{3}LNet, and (j) ADMamba for the first frame of Guadalajara data.

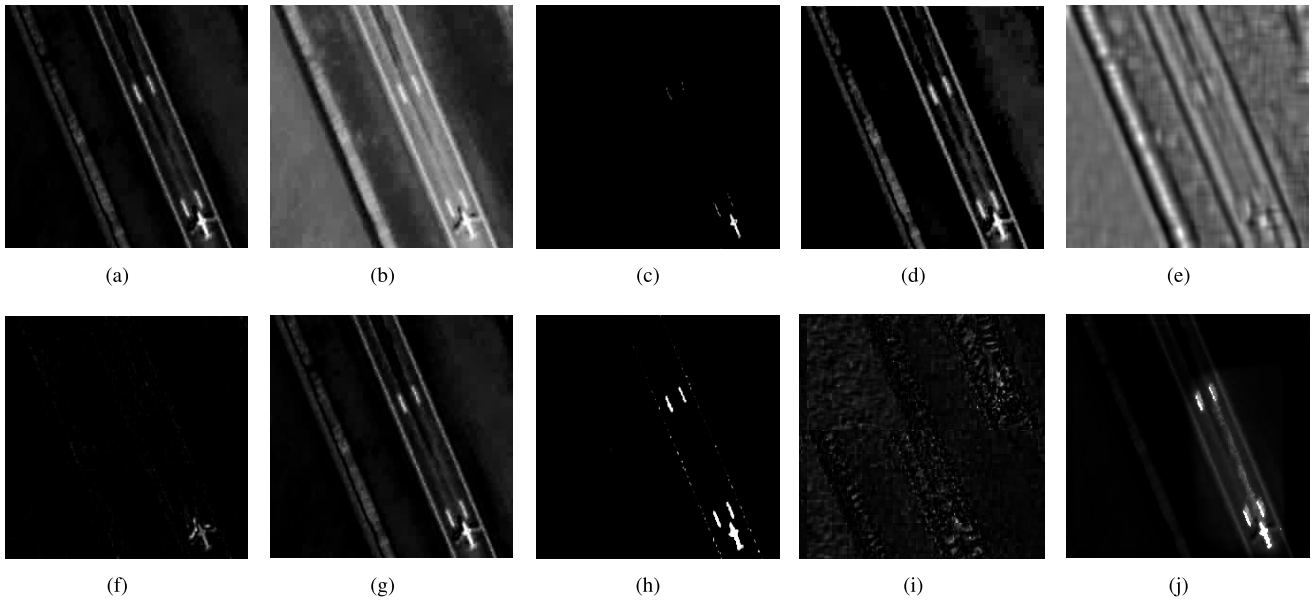


Fig. 11. Detection maps of (a) GRX, (b) LSAD-CR-IDW, (c) SLW-LRRSTO, (d) MLW-LRRSTO, (e) 2S-GLRT, (f) Auto-AD, (g) DSAD, (h) CTAnet, (i) BS^{3}LNet, and (j) ADMamba for the first frame of the Macau data.

demonstrates second-best performance in terms of AUC_{BS} and AUC_{SNPR}, while $AUC_{(D,F)}$ performs well on all datasets, including the Dalian and Marseille datasets where the anomaly proportions are extremely low.

9) Similar to 2S-GLRT, BS^{3}LNet exhibits significant advantages on the Dalian and Marseille data with extremely low anomaly ratios. For instance, on the Dalian data, BS^{3}LNet achieves the optimal performance in four key metrics. On the Marseille dataset, it ranks second in two metrics. As illustrated in Fig. 13, BS^{3}LNet accurately detects ships with strong background suppression effectiveness.

10) In most cases, ADMamba outperforms other methods. As shown in Table I and Figs. 8 and 9, ADMamba achieves the best or second-best results for 4, 5, 6, 4, and 8 metrics across five satellite video datasets. Notably, ADMamba achieves the highest or second highest performance for both $AUC_{(D,F)}$ and AUC_{BS} across all datasets. As seen in Fig. 7, ADMamba's $AUC_{(D,F)}$ and its corresponding standard deviations are the best in most cases. In addition, as demonstrated in Figs. 10–14, the detection maps produced by ADMamba are closer to the ground truth, particularly in terms of AUC_{SNPR}, where ADMamba more clearly highlights anomaly information while effectively suppressing background noise.

11) The computational time of different methods is shown in Table II. As can be seen, GRX is computationally efficient compared to other methods, with around 2.1–2.6 s.

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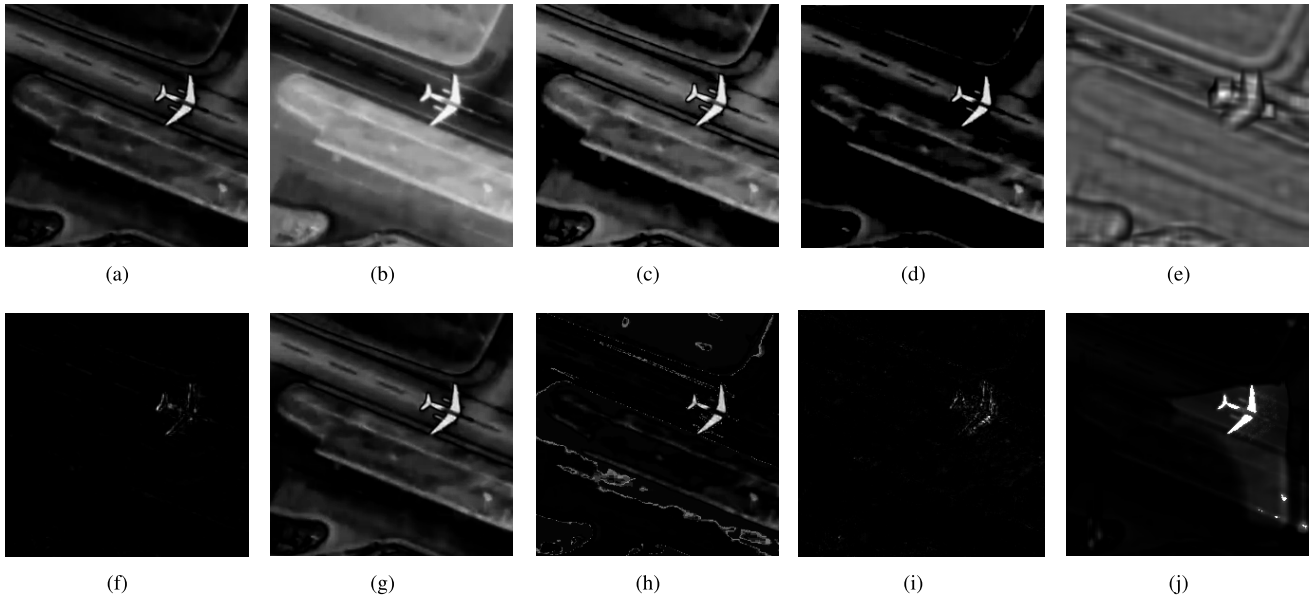


Fig. 12. Detection maps of (a) GRX, (b) LSAD-CR-IDW, (c) SLW-LRRSTO, (d) MLW-LRRSTO, (e) 2S-GLRT, (f) Auto-AD, (g) DSAD, (h) CTAnet, (i) BS^{3}LNet, and (j) ADMamba for the first frame of the Bahrain data.

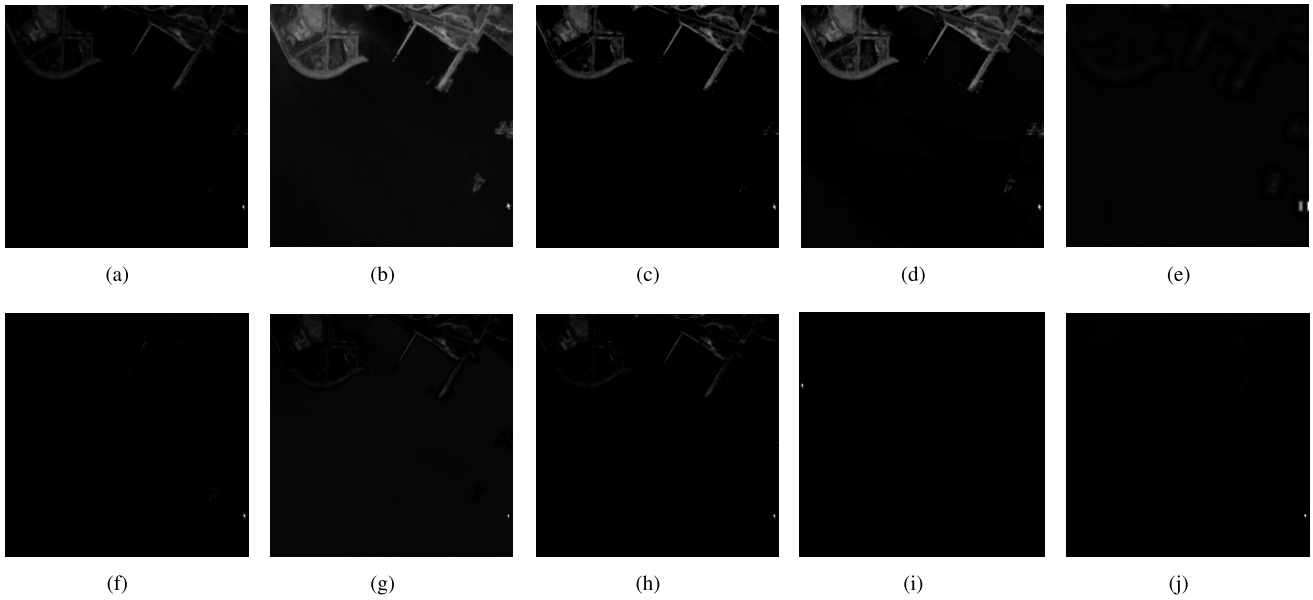


Fig. 13. Detection maps of (a) GRX, (b) LSAD-CR-IDW, (c) SLW-LRRSTO, (d) MLW-LRRSTO, (e) 2S-GLRT, (f) Auto-AD, (g) DSAD, (h) CTAnet, (i) BS^{3}LNet, and (j) ADMamba for the first frame of the Dalian data.

The LSAD-CR-IDW, SLW-LRRSTO, MLW-LRRSTO, and 2S-GLRT algorithms exhibit notably higher computation times, ranging from several hundred (LSAD-CR-IDW) to over a hundred thousand (2S-GLRT). Among the deep-learning-based methods, Auto-AD stands out as the most efficient, with the least time consumption. CTAnet, on the other hand, has the longest training and testing times. ADMamba has a training time of about 17 h but a relatively short testing time of only about 3 s. The total FLOPs and parameters of different deep learning-based methods are compared in Fig. 15. Due to significant differences in the design and training methods of various approaches, to ensure fair comparison, the FLOPs presented in Fig. 15 refer to the total FLOPs consumed for one complete epoch of training on satellite video data with a size of $256 \times 256 \times 64$. The results demonstrate that our model maintains AD performance while keeping its computational cost efficient compared to other deep learning methods, achieving a balance between performance and computational expense for AD of satellite video tasks. In a nutshell, the above-mentioned experimental results and analyses demonstrate the superior performance of the ADMamba method.

E. Ablation Study

In this section, we perform several ablation studies to confirm the benefits of the proposed ADMamba. Specifically, the experiments are conducted using the Guadalajara data from

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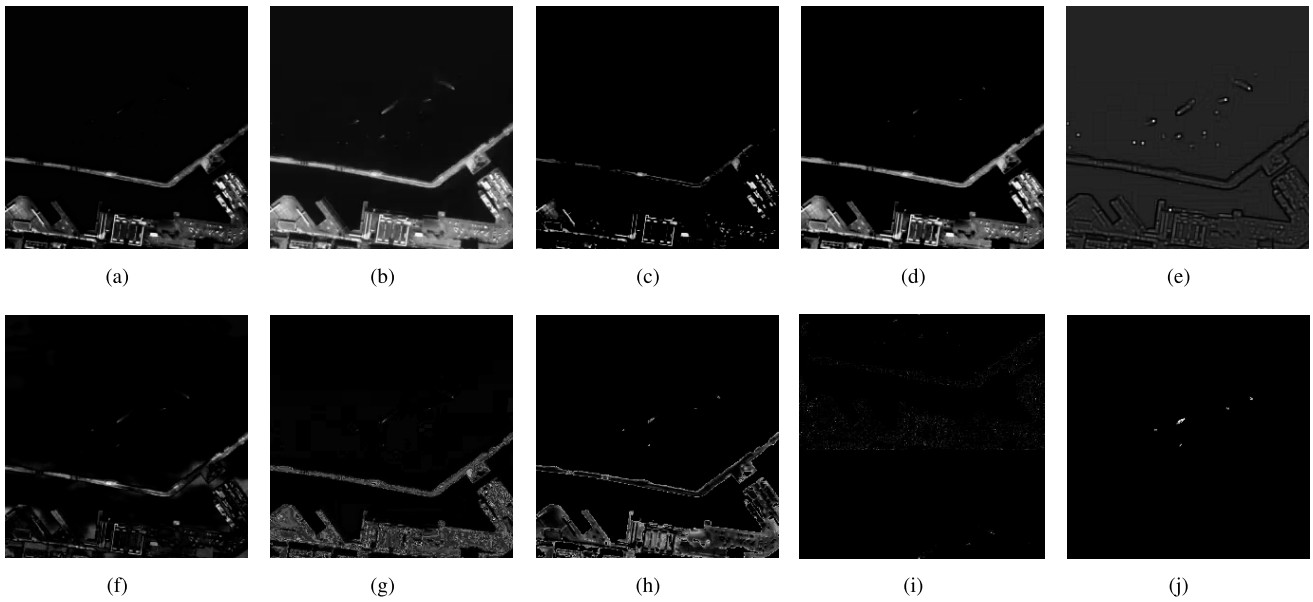


Fig. 14. Detection maps of (a) GRX, (b) LSAD-CR-IDW, (c) SLW-LRRSTO, (d) MLW-LRRSTO, (e) 2S-GLRT, (f) Auto-AD, (g) DSAD, (h) CTAnet, (i) BS^{3}LNet, and (j) ADMamba for the first frame of the Marseille data.

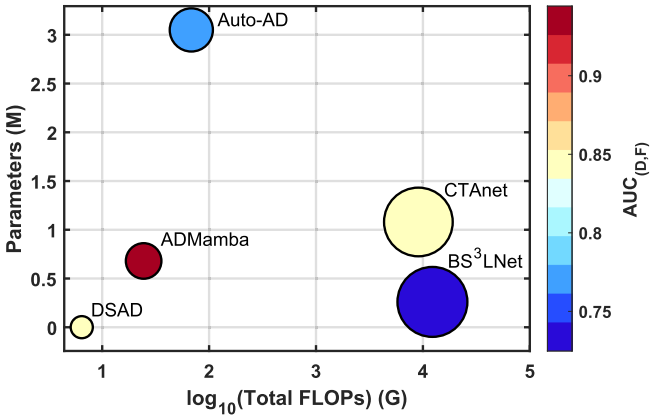


Fig. 15. Comparison of total FLOPs and parameters for the proposed ADMamba and deep learning-based methods on the Guadalajara data.

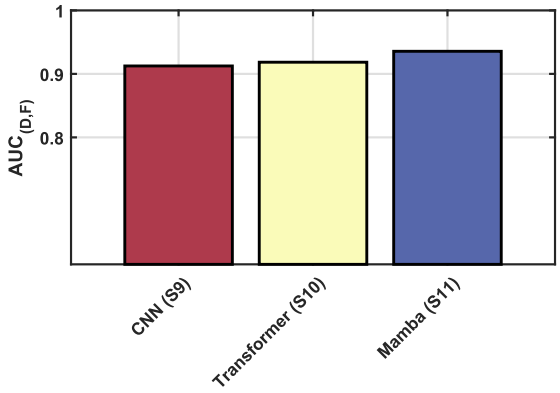


Fig. 16. Comparison of $AUC_{(D,F)}$ for convolutional (***S9***), transformer (***S10***), and Mamba-based (***S11***) backbones on the Guadalajara data.

the Jilin-1 dataset. We gradually remove the clustering-based coarse detection module, the 1-D temporal net, the 2-D spatial net, the 3-D spatio-temporal net, and the Mamba block to evaluate the changes in detection performance (see Table III). The following nine scenarios are considered.

1) *Scenario 1 (S1):* When the clustering-based coarse detection module is removed, only the Mamba-based fine detection module remains in the ADMamba, making the contrastive loss [see (17)] meaningless. To address this, we use the mean square error (mse) to train the network. In this scenario, it is not possible to specifically train the network to minimize the reconstruction error for pseudo-background information while simultaneously maximizing the error for pseudo-anomalous data.
2) *Scenario 2 (S2):* When we remove the 1-D temporal net from the Mamba-based fine detection module, the $\lambda_{1\text{-}D}\mathcal{L}_{1\text{-}D}$ term in the loss function $\mathcal{L}$ is also removed so as to train the network. In addition, the $\|\boldsymbol{V} - \hat{\boldsymbol{V}}_{1\text{-}D}\|^2$ term related to the 1-D temporal net in (23) is removed to obtain the final detection result.
3) *Scenario 3 (S3):* When the 2-D spatial net is removed from the Mamba-based fine detection module, the $\lambda_{2\text{-}D}\mathcal{L}_{2\text{-}D}$ term in the contrastive loss $\mathcal{L}$ and the $\|\boldsymbol{V} - \hat{\boldsymbol{V}}_{2\text{-}D}\|^2$ term associated with the 2-D spatial net are also eliminated to train and test the network.
4) *Scenario 4 (S4):* When the 3-D spatio-temporal net is removed from the Mamba-based fine detection module, both the $\lambda_{3\text{-}D}\mathcal{L}_{3\text{-}D}$ term in the contrastive loss $\mathcal{L}$ and the $\|\boldsymbol{V} - \hat{\boldsymbol{V}}_{3\text{-}D}\|^2$ term related to the 3-D spatio-temporal net are also removed for training and testing the model.
5) *Scenario 5 (S5):* When both the 2-D spatial net and 3-D spatio-temporal net are removed from the ADMamba, leaving only the 1-D temporal net in the Mamba-based fine detection module, all terms related to the 2-D and 3-D-based channels are also removed from the contrastive loss and reconstruction error.
6) *Scenario 6 (S6):* When the 1-D temporal net and the 3-D spatio-temporal net are excluded from the ADMamba, leaving only the 2-D spatial net in the Mamba-based fine detection module, all components related to the

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TABLE III

RESULTS OF THE ABLATION STUDY ASSESSING THE CONTRIBUTION OF EACH COMPONENT IN ADMAMBA, WITH THE BEST-PERFORMING COMPONENT HIGHLIGHTED IN BOLD AND THE SECOND-BEST IN ITALIC UNDERLINE

Different Components	*S1*	*S2*	*S3*	*S4*	*S5*	*S6*	*S7*	*S8*	*S12*
Clustering-based Coarse Detection Module	✗	✓	✓	✓	✓	✓	✓	✓	✓
1D Temporal Net	✓	✗	✓	✓	✓	✗	✗	✓	✓
2D Spatial Net	✓	✓	✗	✓	✗	✓	✗	✓	✓
3D Spatio-temporal Net	✓	✓	✓	✗	✗	✗	✓	✓	✓
Mamba Block	✓	✓	✓	✓	✓	✓	✓	✗	✓
$AUC_{(D,F)}$	0.8506	*0.9380*	0.9286	0.9348	0.7744	0.8802	0.8716	0.9120	**0.9443**

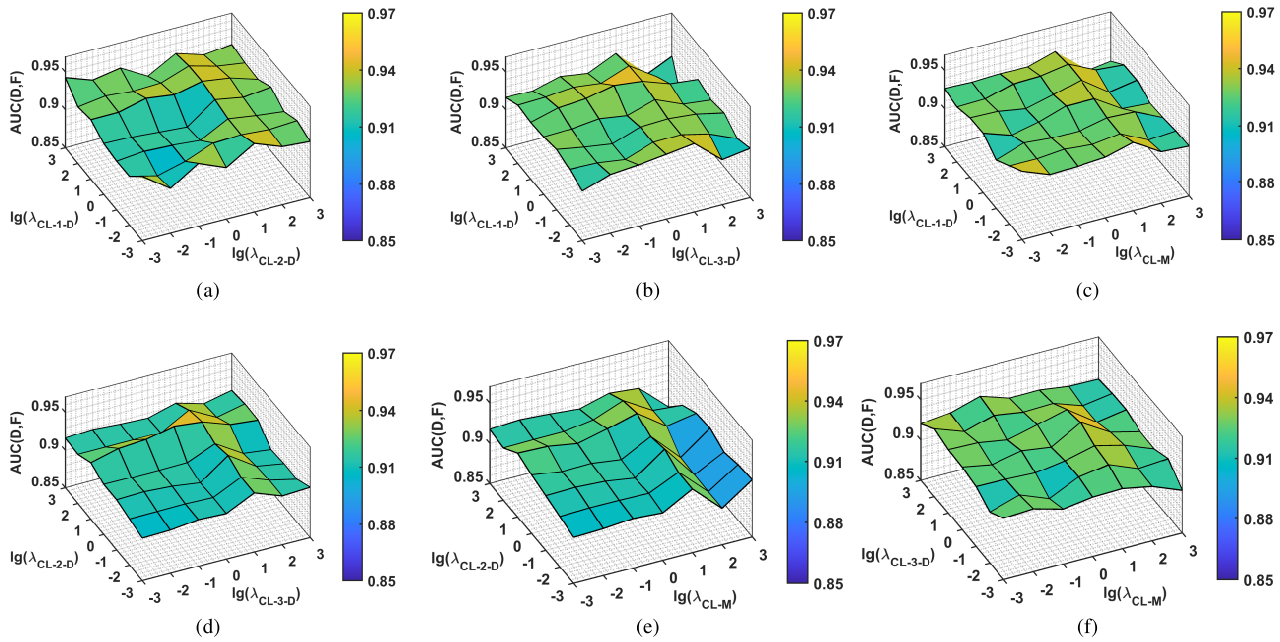


Fig. 17. Impact of the regularization parameters $\lambda_{CL\text{-}1\text{-}D}$, $\lambda_{CL\text{-}2\text{-}D}$, $\lambda_{CL\text{-}3\text{-}D}$, and $\lambda_{CL\text{-}M}$ on the Guadalajara data. Subplots show the joint effects of (a) $\lambda_{CL\text{-}1\text{-}D}$ and $\lambda_{CL\text{-}2\text{-}D}$, (b) $\lambda_{CL\text{-}1\text{-}D}$ and $\lambda_{CL\text{-}3\text{-}D}$, (c) $\lambda_{CL\text{-}1\text{-}D}$ and $\lambda_{CL\text{-}M}$, (d) $\lambda_{CL\text{-}2\text{-}D}$ and $\lambda_{CL\text{-}3\text{-}D}$, (e) $\lambda_{CL\text{-}2\text{-}D}$ and $\lambda_{CL\text{-}M}$, and (f) $\lambda_{CL\text{-}3\text{-}D}$ and $\lambda_{CL\text{-}M}$.

1-D and 3-D channels are also eliminated from both the contrastive loss and the reconstruction error.

7) *Scenario 7 (S7):* When the 1-D temporal net and the 2-D spatial net are removed from the ADMamba, leaving only the 3-D spatio-temporal net in the Mamba-based fine detection module, all terms associated with the 1-D and 2-D channels are also removed from both the contrastive loss and the reconstruction error.
8) *Scenario 8 (S8):* All the Mamba blocks are removed from the Mamba-based fine detection module to assess the contribution of the Mamba blocks to the ADMamba method proposed in this article.
9) *Scenario 9 (S9):* All Mamba blocks are substituted by convolutional blocks.
10) *Scenario 10 (S10):* All ResBlocks, convolutional layers, and Mamba blocks are replaced with transformer blocks.
11) *Scenario 11 (S11):* All ResBlocks and convolutional layers are replaced with Mamba blocks. **S9** to **S11** further examine the performance gain attributable to the use of Mamba.
12) *Scenario 12 (S12):* The whole ADMamba proposed in this article.

The $AUC_{(D,F)}$ values for the methods mentioned above are presented in Table III, where it can be observed that using one, two, three, or four components does not yield as good AD performance as using all components together. For example, in **S1**, the $AUC_{(D,F)}$ decreases by 0.0937. Similarly, for **S2**–**S9**, the detection accuracy shows varying degrees of reduction. **S5** experiences the greatest decline, representing the worst performance among all scenarios. These results indicate that the components within ADMamba are complementary to one another and mutually essential. Moreover, Fig. 16 compares the $AUC_{(D,F)}$ of different backbones on the Guadalajara data. We observe that while **S9** yields the lowest accuracy among the three methods, it still surpasses the second-best method (MLW-LRRSTO) from Table I. The transformer-based backbone (**S10**) achieves better performance than **S9**, while the Mamba-based method performs the best, outperforming **S9** and **S10** by approximately 0.2 in $AUC_{(D,F)}$, respectively. This result further demonstrates the advantage of the Mamba architecture for satellite video AD.

F. Analysis of Parameter Sensitivity

In this section, the sensitivity of nine crucial parameters is discussed, which include the regularization parameters $\lambda_{CL\text{-}1\text{-}D}$,

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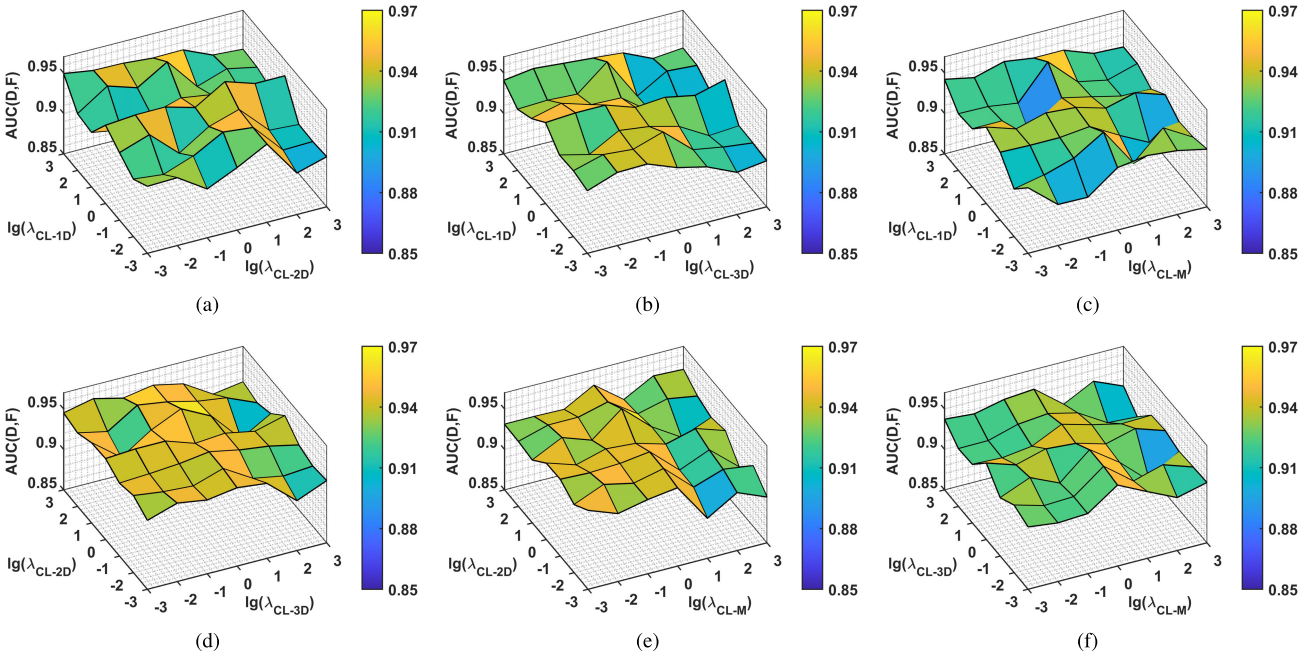


Fig. 18. Impact of the regularization parameters $\lambda_{1\text{-}D}$, $\lambda_{2\text{-}D}$, $\lambda_{3\text{-}D}$, and λ_M on the Guadalajara data. Subplots show the joint effects of (a) $\lambda_{1\text{-}D}$ and $\lambda_{2\text{-}D}$, (b) $\lambda_{1\text{-}D}$ and $\lambda_{3\text{-}D}$, (c) $\lambda_{1\text{-}D}$ and λ_M, (d) $\lambda_{2\text{-}D}$ and $\lambda_{3\text{-}D}$, (e) $\lambda_{2\text{-}D}$ and λ_M, and (f) $\lambda_{3\text{-}D}$ and λ_M.

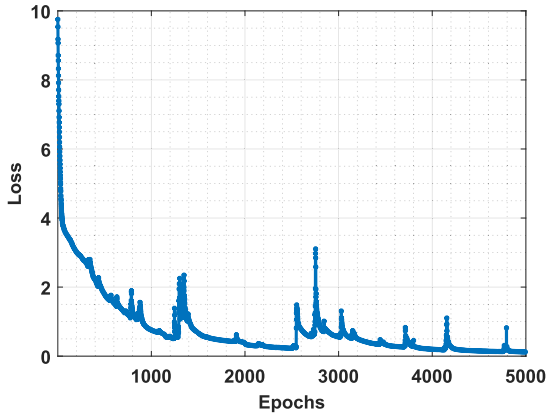


Fig. 19. Impact of the number of epochs on the Guadalajara data.

$\lambda_{CL\text{-}2\text{-}D}$, $\lambda_{CL\text{-}3\text{-}D}$, and $\lambda_{CL\text{-}M}$ shown in (18)–(21), the regularization parameters $\lambda_{1\text{-}D}$, $\lambda_{2\text{-}D}$, $\lambda_{3\text{-}D}$, and λ_M shown in (17), and the number of epochs. First, Fig. 17 illustrates the impact of the regularization parameters $\lambda_{CL\text{-}1\text{-}D}$, $\lambda_{CL\text{-}2\text{-}D}$, $\lambda_{CL\text{-}3\text{-}D}$, and $\lambda_{CL\text{-}M}$ on the Guadalajara data. All the four parameters are selected from $\{0.001, 0.01, 0.1, 1, 10, 100, 1000\}$. It can be observed that the $AUC_{(D,F)}$ fluctuates as the regularization parameters change. When the parameters are generally within the range of 10–1000, the performance of AD is better than that with other values. Therefore, it is reasonable for us to set $\lambda_{CL\text{-}1\text{-}D} = \lambda_{CL\text{-}2\text{-}D} = \lambda_{CL\text{-}3\text{-}D} = \lambda_{CL\text{-}M} = 10$ in the experiment. Second, the influence of the regularization parameters $\lambda_{1\text{-}D}$, $\lambda_{2\text{-}D}$, $\lambda_{3\text{-}D}$, and λ_M is depicted in Fig. 18. These four parameters are also chosen from the set $\{0.001, 0.01, 0.1, 1, 10, 100, 1000\}$. As can be observed, $AUC_{(D,F)}$ also fluctuates with the changes of these four parameters. Moreover, when $\lambda_{1\text{-}D} = \lambda_{2\text{-}D} = \lambda_{3\text{-}D} = \lambda_M = 1$, a satisfactory AD effect can be achieved. Therefore, in the experiment, we set these four parameters to 1. Finally, the influence of epochs is analyzed in Fig. 19, within which the number of epochs varies from 1 to 5000. As is evident from Fig. 19, the loss experiences a rapid decline during the initial approximately 1000 epochs. Subsequently, it fluctuates with a general downward trend until approximately 2500 epochs. There is then a significant spike in loss around 2500 epochs, followed by another decline with some fluctuations. Overall, as the epochs progress toward 5000, the loss gradually approaches a relatively stable and low value. In the experiment, we set the number of epochs to 5000.

IV. Conclusion

In this article, we have proposed a spatio-temporal Mamba-inspired network, namely, ADMamba, for AD of satellite videos. The ADMamba architecture consists of two main modules: a clustering-based coarse detection module and a Mamba-based fine detection module. Compared to existing AD methods, the proposed ADMamba has two main advantages. First, the ADMamba is an unsupervised AD method that processes satellite video information in a coarse-to-fine fashion. The pseudo-coarse background and anomaly samples can be detected by the clustering-based coarse detection module without relying on any prior information about anomalies and background pixels, while the Mamba-based fine detection module refines the coarse detection results by "U"-shape channels. Second, the ADMamba is a novel Mamba-based deep learning method for AD in satellite videos, the 1-D, 2-D, and 3-D Mamba-based architectures can effectively extract and reconstruct the temporal, spatial, and spatio-temporal features of satellite videos. Both quantitative and qualitative evaluations based on five real-world Jilin-1 and OVS-1 satellite videos reveal that the AD outcomes obtained by ADMamba outperform the state-of-the-art methods. The proposed ADMamba method also has limitations. The pseudo-coarse background and anomaly samples may lead to biased learning in the model, which is a major open challenge. Future research will focus on addressing the above limitations, further optimize the

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network structure using architecture search methods, explore the model's capacity to detect diverse types of anomalies, and apply the proposed method to AD in more complex scenarios.

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